PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. under the Panjab University Act VII of 1947—enacted by the Govt. of India)

SYLLABI

AND THE

STRUCTURAL OUTLINE

FOR

B.A./B.Sc. (Honours) Course
Third Year Examination, 2011

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STRUCTURAL OUTLINE OF THE B.A./B.SC. (GENERAL AND HONOURS) DEGREE COURSE IN THE 10+2+3 SYSTEM OF EDUCATION FOR FIRST YEAR 2009 EXAMINATION (ALREADY HELD); SECOND YEAR 2010 (ALREADY HELD); THIRD YEAR 2011 EXAMINATION

The new B.A./B.Sc. (General & Honours) course (at + stage) in the 10+2+3 system of education has been structured keeping in view certain basic guidelines as under :--

1. *(i)* The scheme should provide a broad-based, interrelated and comprehensive programme of study, built upon the + 2 stage, which ultimately may act as a sound foundation for postgraduate education, participation in competitive examinations, and joining professional courses like those in Law, Education, etc.

   (ii) It is so structured and organised that it is in conformity with the needs of the country in the present context. Apart from providing good general education, it should have an applied component as part of the course work so that it creates awareness of the practical problems facing the society and generates employability of the youth in diverse spheres of activity.

   (iii) It is so designed and formulated that it provides flexibility and relative freedom of choice of courses to the students according to their special interests and capabilities.

   (iv) It is based on intensive teacher–student interaction.

2. The B.A./B.Sc. courses have been so structured and revised that they are comparable both in content and quality.

3. To cope with the fast increasing knowledge in each discipline, it has been considered desirable to cover each elective subject in two papers.

The two-paper scheme will enable the teachers to teach the subject in detail and examine the students accordingly. Effort has to be made to improve the level and adequacy of content of each course in such a manner that the total scheme provides distinct upgradation of the existing curricula for the first degree. In the absence of such an upgradation, the increase in the total educational period from 14 to 15 years for the B.A./B.Sc. will be unjustifiable.

Keeping in view the above objectives and guidelines, the following structural outline has been introduced for the B.A./B.Sc. (General) degree course :

Admission to B.A./B.Sc. (General and Honours) course for the second and third year (for the students who have passed First Year under this scheme) for the session 2008-2009 (2009 examination) shall be as hereinafter mentioned.

Note : A person who has passed B.A./B.Sc. 1st year or 2nd year examination from other Universities in India may be allowed to migrate to this University subject to the condition that he shall have to clear the deficient subject but the total number of credits required to be earned shall remain the same.

(ii) STRUCTURAL OUTLINE OF THE B.A./B.SC. (GENERAL/HONOURS) DEGREE COURSE IN THE 10+2+3 SYSTEM OF EDUCATION

B.A. (General)

The B.A. (General) programme of study shall consist of **24** credits, each credit having a value of 100 marks. A subject studied for the whole academic year shall carry 2 credits. All the theory papers and practicals irrespective of their credit value shall be studied throughout the academic year.

The detail of the subjects to be studied shall be as under :

1st Year 2009 Examination (already held)

COMPULSORY

(a)	Punjabi – Two Papers	1 Credit	These papers constitute one subject in each year.
	OR		
	*History & Culture of Punjab –One Paper	1 Credit	
(b)	English	1 Credit	

ELECTIVE

+ Any three elective subjects of two credits each including languages and Elective vocational subject (To be studied in selected colleges)	6 Credits
Total :	**8 Credits**

Environment Education : 50 marks

2nd Year 2010 Examination (already held)

Same as in the 1st year

3rd Year 2011 Examination

Same as in the 1st year and 2nd year examination.

Note :

1. In subjects having practicals, the theory papers and practicals together will be of 2 credit value. The candidate will have to pass in theory and practical/s separately.

2. Each paper of one credit shall be allocated 3 hours of teaching per week during each academic year. However, for subjects having practicals, three hours of teaching will be allocated to each theory paper and two hours for a practical per week or as per requirement of the subject concerned.

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* For B.A./B.Sc. History and Culture of Punjab is allowed as an option in lieu of Punjabi (compulsory) to those students who had already offered earlier in the First Year of 2009 examination.

3. B. A. (General) degree holders shall be eligible for admission to the Master's course in any of the elective subjects studied by them during all the three years of the programme of study, earning six credits in each, provided they fulfil the eligibility conditions.

4. A student would offer any science subject, including Mathematics, only if he has passed that subject in the qualifying examination or qualifies in the subject as a deficient/additional subject from the concerned Board/University/Council in the Supplementary Examination subsequent to the admission.

Provided further that a student can offer :

(a) Statistics only if he takes up Mathematics.

(b) Applied Statistics only if he takes up other subject(s) excluding Mathematics.

B.Sc. (General)

The B.Sc. (General) programme of study shall consist of **20** credits, each credit having a value of 100 marks. A subject studied for the whole academic year shall carry 2 credits. All the theory papers and practicals irrespective of their credit value shall be studied throughout the academic year.

The detail of the subjects to be studied shall be as under :

1st Year 2009 Examination (already held)

Punjabi/History & Culture of Punjab	1 Credit
+ Three Elective Subjects of 2 credits each including Elective Vocational subject (To be studied in selected colleges)	6 Credits
Total :	**7 Credits**

Environment Education : 50 marks

2nd Year 2010 Examination (already held)

English	1 Credit
+ Three Elective Subjects of 2 credits each including Elective Vocational subject (To be studied in selected colleges) (the same as in the 1st year)	6 Credits
Total :	**7 Credits**

3rd Year 2011 Examination

Three Elective Subjects of 2 credits each (the same as in the 1st and 2nd Year)	6 Credits
Total :	**20 Credits**

(iv) STRUCTURAL OUTLINE OF THE B.A./B.SC. (GENERAL/HONOURS) DEGREE COURSE IN THE 10+2+3 SYSTEM OF EDUCATION

Note : 1. The two credits allocated to an elective subject, to be studied in an academic year as spelled out above, shall be covered in two theory papers and a practical or practicals as per requirements of the subject concerned.

2. In the instructional process, it is desireable to strike a balance in the teaching and learning strategies. A close interaction between the teacher and the student is necessary to achieve the above objectives.

3. Each elective subject of 2 credits for the B.Sc. shall be allocated five hours of teaching and four hours of practical work per week during the course of the academic year or as per requirements of the subject concerned.

4. B.Sc. (General) degree holders shall be eligible for admission to the M.Sc. course in any of the three elective subjects which they have studied during all the three years of the degree course, earning 6 credits in each, provided the candidate fulfils the eligibility conditions.

B.A./B.Sc. Honours

In addition to the main objectives listed in the preamble to the B.A./B.Sc. (General) degree course, the Honours Course seeks to provide course work of advanced nature in one of the elective subjects already being studied from 1st year onwards by a student. Both in content and quality, it would be of a level as gives adequate specialized knowledge in a subject even to make it a culminating point in the programme of study if a student so chooses. Or else, it should provide strong, additional foundation for pursuing the subject at the M.A./M.Sc. level.

Keeping in view the objectives stated in the preamble to the B.A./B.Sc. (General) degree course and that enunciated above, the structural outline for the Honours degree course shall be as under :

B.A. (Honours)

The B.A. Honours Course shall carry 28 credits (each credit carrying the equivalence of 100 marks). Apart from the structure of the programme of study prescribed for the B.A. (General) degree course, credit value of the subjects/papers and hours of teaching allocated to them as already stated in the scheme, the Honours Course shall include four papers of one credit each of an advanced nature in one of the elective subjects a student shall study in all the three years of the degree course.

A student may offer Honours in Second Year in any one of the elective subjects, to be studied by him in all the three years of the course.

Provided that he has obtained at least 50 % marks in the subject of Honours in the First Year of the B.A. (General) Course.

For the B.A. (Honours) Course, the distribution of work during the three years will be as under :--

1st Year : The same as in B.A. (General). (8 credits)

2nd Year : The same as in 2nd year of B.A. (General). In addition, there shall be two advanced papers of one credit each in the subject in which he seeks to get Honour degree.

(8+2 = 10 credits)

3rd Year : The same as in 3rd year of B.A. (General). In addition, there shall be two advanced papers of one credit each in the Honours subject.

(8+2 = 10 credits)

Total Credits : 28

A candidate eligible to appear as a private candidate in B.A. examination may also offer Honours papers in any of the elective subjects offered for B.A. (General) in all three years provided he fulfils the requirement of the Regulation.

B.Sc. (Honours)

The B.Sc. (Honours) programme of study shall consist of 24 credits, each credit having a value of 100 marks. A subject studied for the whole academic year shall carry 2 credits. All the theory papers and practicals irrespective of their credit value shall be studied throughout the academic year.

A student may offer Honours in any one of the elective subjects to be studied by him in all the three years of the course provided he has obtained at least 50% marks in the subject concerned in the first year examination of the B.Sc. (General) course.

Of the 24 credits, each student shall offer courses in each year as under :--

1st Year : The same as for First Year of B.Sc. (General). (7 credits)

2nd Year : The same as for 2nd Year of B.Sc. (General). In addition, there shall be two advanced papers of one credit each in the subject in which he seeks to get Honours degree.

(7+2 = 9 credits)

3rd Year : The same as for 3rd year of B.Sc. (General). In addition, there shall be two advanced papers of one credit each in the Honours subject.

(6+2 = 8 credits)

Total Credits : 24

Both in B.A. Honours and B.Sc. Honours Courses, each paper of one credit shall be allocated 3 hours of teaching, or as per requirements of the subject concerned.

Note : Instruction through audio and/or video cassettes may form a part of Languages course.

GUIDELINES REGARDING CONTINUOUS ASSESSMENT OF REGULAR STUDENTS OF B.A./B.SC./B.COM. /B.C.A. COURSES

IMPORTANT NOTE

(i) In order to incorporate an element of continuous assessment of students, the Colleges will conduct two mandatory House Tests in theory papers – one in the month of September/October and the other in December/January every year.

(ii) *(a)* For September Test, there will be only one paper of one hour's duration in each subject, and for December Test, there will be paper/s on the pattern of annual examination conducted by the University.

There will be a Special Test for those students who could not fulfil the conditions of eligibility. It will not be held to provide an opportunity to all students to improve their earlier score. Those students who are exempted by the Principal of the College from appearing in the House Test/s in September and/or December/January will also be allowed to appear in the Special Test; this Test will determine their eligibility for admission to the examination as well as their score for Internal Assessment.

(b) With a view to meet the grievance of students, if any, on account of scores obtained by them, the answer-books will be shown to them. Difference of opinion on the issue, if any, will be sorted out with the help of respective Heads of departments as well as the Principal of the College.

(iii) Whereas the September House Test will carry weightage of 40 per cent, the December House Test will have weightage of 60 per cent in each subject/paper. The total weightage for both the Tests taken together shall be10 per cent of the total marks in each theory subject/paper. The weightage of 10 per cent marks shall be added to each paper of B.A./B.Sc./B.Com./B.C.A. I, II and III Year which will, henceforth, carry weightage of maximum marks allotted to each paper. A candidate will have to pass in theory and practical/s separately.

(iv) The record of marks secured by the students in the two House Tests will be sent by the respective Colleges so as to reach the office of Controller of Examinations latest by 15th March, failing which the result of the students shall be shown as 'RLA' and the entire responsibility for this would lie with the Principal/s of the College/s.

(v) The Colleges will continue to forward the internal assessment of the students for Practicals. Projects and similar other activities, wherever applicable, to the Controller of Examinations, as usual, so as to reach his office latest by 15th March.

SPECIAL NOTE :

(i) Each theory question paper will be set out of the marks allotted to each theory paper and 10% marks of the maximum marks of each paper will be internal assessment.

(ii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.

(iii) It will not be mandatory for the students to separately pass in the internal assessment.

PANJAB UNIVERSITY, CHANDIGARH

OUTLINES OF TESTS, SYLLABI AND COURSES OF READING IN THE HONOURS SUBJECTS OF THE B.A./B.Sc. THIRD YEAR (THREE YEAR DEGREE COURSE) EXAMINATION, 2011

ENGLISH

B.A. (HONOURS) THIRD YEAR EXAMINATION, 2011

Paper III : 20TH CENTURY LITERATURE

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hrs.

Texts :

1.	G.B. Shaw	:	*St. Joan*, Ed. A.C. Ward, Orient Longman, Hyderabad.
2.	George Orwell	:	*Animal Farm,* Kalyani Classics, New Delhi.
3.	Joseph Conrad	:	*Heart of Darkness.*
4.	W.B. Yeats	:	Yeats Poems : ed. by Norman Jeffares, Kalyani Pubs., N. Delhi : (i) "Down by the Salley Gardens". (ii) "The Song of Wandering Aengus". (iii) "Lake Isle of Innisfree".
5.	T.S. Eliot	:	*'Portrait of a Lady'*, from *Selected Poems* ed. Manju Jain, OUP, Delhi.

Testing :

1.	One question on age/movements with internal choice.	15 marks
2.	One question with reference to the context. Passages to be set from Nos. 1, 4 and 5. Students will attempt two out of three passages.	7 ½ × 2 =15 marks
3.	One essay-type question with internal choice on each of the five writers and their texts. Students will attempt three questions.	20 × 3 = 60 marks

Paper IV : INDIAN WRITING IN ENGLISH

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hrs.

Texts :

1.	Tagore	:	*Hungry Stones and Other Stories,* Rupa Classics, N. Delhi.
2.	Mulk Raj Anand	:	*Untouchable*, Orient Pubs., Hyderabad.
3.	Raja Rao	:	*Kanthapura,* OUP, N. Delhi.
4.	Salman Rushdie	:	*Haroun and the Sea of Stories,* Penguin, N. Delhi.
5.	Nissim Ezekiel	:	"Night of the Scorpion," "Goodbye Party for Miss Pushpa T. S.".
6.	Kamala Das	:	"The Invitation," " A Hot Noon at Malabar", My Grandmother's House" (10 Twentieth Century Indian Poets, ed. R. Parthasarthy, OUP, Delhi).

Testing :

One question, with internal choice, on each of the writers will be asked. Students will attempt any five questions. Each question will carry 18 marks.

संस्कृत

बी. ए. (आनर्स) तृतीयवर्ष परीक्षा, 2011

तृतीय प्रश्न–पत्र पूर्णांकः 90+10 =100 अंक

समय : 3 घण्टे

उद्देश्यः–

विद्यार्थियों को काव्य के स्वरूप, प्रयोजन इत्यादि से अवगत करना । इसी के साथ ही पाणिनीय सूत्रों के माध्यम से कारक तथा स्त्री प्रत्ययों के अध्ययनोपरान्त संस्कृत लेखन के प्रति उत्साहित करना ।

(i) साहित्यदर्पण–प्रथम परिच्छेद 40 अंक

(ii) लघुसिद्धान्तकौमुदी– 30 अंक

(क) विभक्त्यर्थ (कारक–प्रकरण) 15 अंक

(ख) स्त्री–प्रत्यय 15 अंक

(iii) संस्कृत में निबन्ध (मात्र निम्नलिखित विषयों पर) 20 अंक

संस्कृत भाषा, परोपकार, गीता

वेदानां महत्त्वम्, आदर्श–छात्र : ।

सूचना–अध्यापन समय– 6 घण्टे प्रतिसप्ताह

चतुर्थ प्रश्न–पत्र पूर्णांकः 90+10 =100 अंक

समय : 3 घण्टे

उद्देश्यः–

उपनिषद् प्रतिपादित नचिकेता एवं यम के संवाद द्वारा विद्यार्थियों को रहस्यमय परमतत्त्व से परिचित कराना इसके अतिरिक्त चतुर्वेद, ब्राह्मण तथा उपनिषद् ग्रन्थों के विषय–वस्तु का अध्ययन करवाना।

(खण्ड अ) उपनिषद्–साहित्य

निम्न पाठ्यक्रम निर्धारित हैं –

(i) कठोपनिषद्–अनुवाद एवं व्याख्या (6 में से 4 सन्दर्भ) 4×10 = 40 अंक

(ii) कठोपनिषद् से सम्बद्ध प्रश्न (2 में से 1 का उत्तर) 1×20 = 20 अंक

(खण्ड ब) वैदिक साहित्य का इतिहास

निम्न पाठ्यक्रम निर्धारित हैं –

1. सहिंताएँ–विषय वस्तु 30 अंक
2. ब्राह्मणग्रन्थ–विषय वस्तु
3. उपनिषद् साहित्य–विषय वस्तु (चार में से दो के उत्तर पूछे जायें)

सूचना–अध्यापन समय–6 घण्टे प्रतिसप्ताह

Books Recommended :

1. Sanskrit Drama : A.B. Keith.

2. Classical Sanskrit Literature : A.B. Keith.

3. A History of Sanskrit Literature : A.A. Macdone.

4. A History of Indian Literature, Vol. 1 : M. Vintenitz

5. History of Sanskrit Literature by V. Varadachari.

6. संस्कृत साहित्य का इतिहास : बलदेव उपाध्याय

7. वैदिक साहित्य और संस्कृतिः बलदेव उपाध्याय

8. वैदिक साहित्य एक विवेचन : डॉ. वेद प्रकाश उपाध्याय

9. संस्कृत साहित्य की रूपरेखा : चन्द्रशेखर पाण्डेय

Note : All the books are available at the Chaukhamba Sanskrit Series, Chaukhamba, Varanasi and Motilal Banarasidas, Jawahar Nagar, Bunglow Road, Delhi.

PERSIAN

B.A. (HONOURS) THIRD YEAR EXAMINATION, 2011

PAPER-III : HISTORY OF PERSIAN LITERATURE
SAFIVID PERIOD

100 Marks

Time : 3 Hrs.

Books Recommended :

1. History of Persian Literature by Dr. Razazadeh Shafaq.
2. History of Persian Literature by Dr. Brown Vol. III.

PAPER IV : MODERN PERSIAN

100 Marks

Time : 3 Hrs.

1. **Prose :** 50 Marks

   (i) The following three short stories of Saeed-e-Nafisi :

   (a) Azan-e-Maghrib.

   (b) Khana-ye-Pidari.

   (c) Janayat-e-man.

   (ii) The following three short stories of Mohammed Heijazi :

   (a) Khud Kushi.

   (b) Eedi.

   (c) Mah-e-man.

Book Prescribed :

Nasib-e-Jadid-e-Farsi, Jayyed Press, Ballimaran, Delhi-6.

2. **Poetry :** 50 Marks

Quasidan

1. Namund Namah by Bahar
   "Dar Sher Band Mehro-Wafa Dilbari Namund".
2. Ghazalliyat-E-Parveen Etisami by Mehdi Hameedi.
3. Qataat by Mehdi Hameedi.

Books Recommended :

1. Barguzida-ye-She'r-e-Farsi-ye-Mu' aasir Part I by Dr. Muneeb-al-Rehaman.
2. Edara-ye-Ulam-e-Islami, Danish gah-e-Islami, Aligarh.

ARABIC

B.A. (HONOURS) THIRD YEAR EXAMINATION, 2011

(Two Specialised Papers)

PAPER- III : HISTORY OF ISLAM–UMAYYED PERIOD

100 Marks

Time : 3 Hrs.

Books Recommended :

(i) History of Saracens by Ameer Ali.

(ii) Tarikh-ul-Ummat Islam by Jai Raj Puri.

(iii) Tarikh-e-Islam by Prof. K.A. Fariq.

PAPER-IV : MODERN LITERARY HISTORY OF THE ARABS FROM 1798 A.D. TO 1932 A.D.

100 Marks

Time : 3 Hrs.

Books Recommended :

(i) Adab-ul-Arab by Hassan-ul-Ziyyat.
(All books can be had from Kutub Khana-e-Rashidiyya, Urdu Bazar, Jama Masjid, Delhi-6).

(ii) Al Adab Arabi al Moaser by Shoaqui Zef.

FRENCH

B.A. (HONOURS) THIRD YEAR EXAMINATION, 2011

Paper : III : HISTORY OF FRENCH LITTERATURE

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hrs.

Introduction to History of French Literature from 17th–20th century.

Questions should be asked strictly from the prescribed text book with

1.	Two questions of 15 marks each.	30 Marks
2	Three questions of 10 marks each.	30 Marks
3.	Fill in the blanks or multiple choice questions with maximum of 2-3 marks each.	30 Marks

Courses of Reading :

Mèmento de Littérature française, by Hèlène, Potelet, Publication Hatier, Profil Littèrature. Only from 17-20 centuries to be studied.

Note : 1. The latest syllabus should be followed.

2. Choice should be given in questions.

Reference Books :

1. Collection littéraire—Moyen age – 20eme siécle -Lagarde et Michard, Bordas.

2. Recueil de Textes littéraires française. Moyen Age -20eme siécle-Chassang et Senninger.

3. Manual des études françaises.
Moyen âge -20eme siècle – Castex et. P. Surer, Hachette.

Paper : IV : POETRY AND VIVA-VOCE

Max. Marks	:	100
Theory Viva-Voce	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hrs.

Poetry : Marks : 70

Comprehension, explanation, literary, appreciation and criticism of the poems to be studied.

The following 10 poems to be studied :

1. Pour Hèlène-quand vous serez bien vieillie—Pierre Ronsard - French Poetry for students –A.W. Bains.
2. Le Savetier et la financier-La Fontaine, French Poetry for students –A.W. Bains.
3. Le, Rat qui s'est retirè du monde –La Fontaine –French Poetry for students –A.W. Bains.
4. L'isolement –Lamartine –French Poetry for students –A.W. Bains.
5. Puisque j' ai mis ma lèvre –V Hugo French Poetry for students –A.W. Bains.
6. La mort du Loup –A De vigny –French Poetry for students –A.W. Bains.
7. Recueillement-Baudelaire –French Poetry for students –A.W. Bains.
8. Page d'écriture –J. Prevert –Paroles.
9. Si tu t' imagines-Raymond Queneau-L'Instant Fatal.
10. Non r êve familier–P. Verlaine –collection littéraire. Lagarde et Michard. XIX siécle.

Note :
1. The latest syllabus should be strictly followed.
2. Choice should be given in questions.

Text Books for Reference :

1. French Poetry for students by A.W. Bains.
2. Paroles edition Folio by Jacques Prevert.
3. Le francais et la Vie-Tome III G. Mauger/M.Bruezier e.
4. Collection Litteraire XIX Siecle-Nouveaux classique Larousse.
5. Poems choisis –A de. Musset –eidtion –Nouveaux classique Larousse.

Viva-Voce 20 marks

1. Dictation of an unseen passage of about 200 words. : 10 Marks

2. Oral questions based on the dictation passage and also general questions : 10 Marks

Note : Questions in all the papers shall be asked and answered in French only.

हिन्दी

बी. ए. (आनर्स) तृतीय वर्ष परीक्षा, 2011

तृतीय पाठ्यचर्या

पूर्णांक : 90+10 = 100

समय : 3 घण्टे

1. काव्य–नाटकः एक कण्ठ विषपायी–दुष्यंत कुमार, लोक भारती प्रकाशन, इलाहाबाद।
2. कहानियाँः आधार चयन–भीष्म साहनी, आधार पब्लिकेशन, पंचकूला।
3. संस्मरणः अतीत के चलचित्र–महादेवी वर्मा, राधाकृष्ण प्रकाशन, नई दिल्ली।

नोट :

(क) इस पाBयचर्या में कुल मिलाकर तीनों काव्य कृतियों में से 10–10 अंकों के छः व्याख्यांश पूछे जायेंगे। छात्रों को केवल तीन की ही संदर्भ एवं काव्य सौन्दर्य सहित व्याख्या करनी होगी।

(ख) तीन कृतियों में से कम से कम आठ समीक्षात्मक प्रश्न पूछे जाएंगे जिनमें से छात्रों को 15–15 अंकों के चार प्रश्नों के उत्तर देने होंगे।

विशेष : छात्रों को प्रत्येक कृति में से कम से कम एक प्रश्न अवश्य करना होगा।

(ग) **आंतरिक मूल्यांकन** अंक : 10

सहायक पुस्तकें :

1. अंधा युग : पथ और प्रदर्शन, जयदेव तनेजा, वाणी प्रकाशन, नई दिल्ली।
2. अन्धा युग और भारती के अन्य नाट्य प्रयोग : जयदेव तनेजा, नचिकेता प्रकाशन, नई दिल्ली।
3. अन्धा युग : एक विवेचन, डॉ. हरीश्चन्द्र वर्मा, नव भारती शंकर प्रतिष्ठान, नई दिल्ली।
4. हिन्दी कहानी अन्तरंग परिचय : रामदरश मिश्र, नेशनल पब्लिकेशन, नई दिल्ली।
5. हिन्दी कहानी अपनी जुबानी : डॉ. इन्द्रनाथ मदान, संभावना पब्लिकेशन, हापुड़ ।

चतुर्थ पाठ्यचर्या **पूर्णांक : 90+10 = 100**

समय : 3 घण्टे

(अ) काव्यशास्त्र **अंक : 60**

1. छंद : भुजंगप्रयात, मालिनी, मन्दाक्रांता, शिखरिणी, अहीर, मानव, बरवै अंकः 20

2. अलंकारः अन्योक्ति, तुल्ययोगिता, उल्लेख, व्याज, अनन्वय, विभावना, अंकः 20

   विशेषोक्ति, दृष्टान्त, प्रतीप, उपमेयोपमा, व्यतिरेक

3. (क) शब्द–शक्तियाँ– परिभाषा और उदाहरण अंकः 20

   (ख) बिम्ब– परिभाषा और स्वरूप

   (ग) रस– परिचय, प्रकार, लक्षण और उदाहरण

   (घ) प्रतीक–परिभाषा, स्वरूप

विशेष :

1. छात्रों को पाँच अंकों के 4 छंदों और पाँच–पाँच अंकों के 4 अलंकारों के लक्षण तथा उदाहरण लिखने होंगे। प्राश्निक को दुगना विकल्प देना होगा।

2. खंड तीन से संबंधित चारों उपखंडों (क, ख, ग, घ) में से कुल मिलाकर 10–10 अंकों के चार प्रश्न पूछे जायेंगे जिनमें से छात्रों को किन्हीं दो के उत्तर देने होंगे।

(आ) भाषा : **अंकः 30**

1. हिन्दी की प्रमुख बोलियां– क्षेत्र और विशेषताएं (विशेषतः ब्रज, अवधी का परिचय)

2. मानक हिन्दी की संरचना– शब्द संरचना, रूप–संरचना, वाक्य–संरचना।

3. (क) राजभाषा हिन्दी का परिचय तथा वर्तमान स्थिति।.

   (ख) देवनागरी लिपि का अनुमोदित मानक रूप।

   विशेष : 'आ'' खंड में से 15–15 अंकों के कुल चार प्रश्न पूछे जाएंगे, जिनमें से छात्रों को दो के उत्तर देने होंगे।

4. **आंतरिक मूल्यांकन** **अंकः 10**

सहायक पुस्तक :

काव्य के तत्त्व : आचार्य देवेन्द्रनाथ शर्मा, लोकभारती प्रकाशन, इलाहाबाद–1

ਪੰਜਾਬੀ

ਬੀ. ਏ. (ਆਨਰਜ਼) ਭਾਗ ਤੀਜਾ 2●11 ਦੇ ਇਮਤਿਹਾਨ ਲਈ

ਪੇਪਰ ਤੀਜਾ : ਪੰਜਾਬੀ ਵਾਰਤਕ ਕੁੱਲ ਅੰਕ : 1●●

ਲਿਖਤੀ : 9●

ਇੰਟਰਨਲ ਅਸੈਸਮੈਂਟ : 1●

ਸਮਾਂ : 3 ਘੰਟੇ

ਪਾਠਕ੍ਰਮ/ਕੋਰਸ

1. ***ਪੁਰਾਤਨ ਜਨਮ ਸਾਖੀ,*** ਭਾਈ ਵੀਰ ਸਿੰਘ, ਪਬਲੀਕੇਸ਼ਨ ਬਿਊਰੋ, ਪੰਜਾਬ ਯੂਨੀਵਰਸਿਟੀ, ਚੰਡੀਗੜ੍ਹ

2. ***ਘਰ ਦਾ ਪਿਆਰ ਤੇ ਹੋਰ ਲੇਖ,*** ਪ੍ਰਿੰ. ਤੇਜਾ ਸਿੰਘ, ਪੰਜਾਬੀ ਸਾਹਿਤ ਪ੍ਰਕਾਸ਼ਨ, ਬਜ਼ਾਰ ਮਾਈ ਸੇਵਾਂ, ਅੰਮ੍ਰਿਤਸਰ

3. ***ਖਾਨਾ ਬਦੋਸ਼,*** ਅਜੀਤ ਕੌਰ, ਨਵਯੁਗ ਪਬਲਿਸ਼ਰਜ਼, ਚਾਂਦਨੀ ਚੌਕ, ਦਿੱਲੀ

ਯੂਨਿਟ ਅਤੇ ਥੀਮ

1. ਵਿਹਾਰਕ ਆਲੋਚਨਾ (ਹਰੇਕ ਪੁਸਤਕ ਵਿਚੋਂ ਇਕ ਪ੍ਰਸ਼ਨ ਲਾਜ਼ਮੀ) 2×1●=2● ਅੰਕ
(ਚਾਰ ਵਿਚੋਂ ਦੋ)

2. ਤਿੰਨਾਂ ਪੁਸਤਕਾਂ ਵਿਚੋਂ ਛੋਟੇ/ਸੰਖੇਪ ਪ੍ਰਸ਼ਨ (ਹਰੇਕ ਪੁਸਤਕ ਵਿਚੋਂ ਦੋ ਪ੍ਰਸ਼ਨ ਲਾਜ਼ਮੀ) 5×5=25 ਅੰਕ

(ਅੱਠਾਂ ਵਿਚੋਂ ਪੰਜ)

3. ਤਿੰਨਾ ਪੁਸਤਕਾਂ ਦੇ ਛੇ ਪ੍ਰਸ਼ਨਾਂ ਵਿਚੋਂ 3 ਦੇ ਉੱਤਰ (ਹਰੇਕ ਪੁਸਤਕ ਵਿਚੋਂ ਦੋ ਪ੍ਰਸ਼ਨ ਲਾਜ਼ਮੀ ਪੁੱਛੇ ਜਾਣਗੇ) 3×15=45 ਅੰਕ

ਪੇਪਰ ਚੌਥਾ : ਪੰਜਾਬੀ ਸਾਹਿਤ ਦੇ ਪ੍ਰਮੁੱਖ ਝੁਕਾਅ ਅਤੇ ਰੂਪ

ਕੁੱਲ ਅੰਕ : 1●●
ਲਿਖਤੀ : 9●
ਇੰਟਰਨਲ ਅਸੈਸਮੈਂਟ : 1●
ਸਮਾਂ : 3 ਘੰਟੇ

ਪਾਠਕ੍ਰਮ/ਕੋਰਸ

1. ਮੱਧਕਾਲੀ ਪੰਜਾਬੀ ਕਾਵਿ ਦੇ ਝੁਕਾਅ — 45 ਅੰਕ
ਗੁਰਮਤ ਕਾਵਿ, ਸੂਫ਼ੀ ਕਾਵਿ, ਕਿੱਸਾ ਕਾਵਿ, ਬੀਰ ਕਾਵਿ

2. ਆਧੁਨਿਕ ਪੰਜਾਬੀ ਸਾਹਿਤ ਦੇ ਪ੍ਰਮੁੱਖ ਰੂਪ — 45 ਅੰਕ
ਪੰਜਾਬੀ ਕਵਿਤਾ, ਗਲਪ (ਨਾਵਲ, ਕਹਾਣੀ), ਨਾਟਕ ਤੇ ਵਾਰਤਕ ਦਾ ਵਿਕਾਸ

ਯੂਨਿਟ ਅਤੇ ਥੀਮ

ਪ੍ਰਸ਼ਨ ਪੱਤਰ ਦੇ ਦੋ ਭਾਗ ਹੋਣਗੇ। ਹਰੇਕ ਭਾਗ ਵਿਚੋਂ ਪੰਜ-ਪੰਜ ਪ੍ਰਸ਼ਨ ਪਾਏ ਜਾਣਗੇ ਅਤੇ ਵਿਦਿਆਰਥੀ ਨੂੰ ਹਰੇਕ ਭਾਗ ਵਿਚੋਂ ਦੋ-ਦੋ ਪ੍ਰਸ਼ਨਾਂ ਦੇ ਉੱਤਰ ਦੇਣ ਲਈ ਕਿਹਾ ਜਾਵੇਗਾ ਅਰਥਾਤ ਚਾਰ ਪ੍ਰਸ਼ਨ ਕਰਨੇ ਹੋਣਗੇ ਤੇ ਹਰੇਕ ਪ੍ਰਸ਼ਨ ਦੇ ਨੰਬਰ ਬਰਾਬਰ (22½) ਹੋਣਗੇ।

URDU

B.A. (HONOURS) THIRD YEAR EXAMINATION, 2011

PAPER-III : PROSE

Max. Marks : 100
Time : 3 Hrs.

Book Prescribed

Ghubar-e-Khatir by Maulana Abdul Kalam Azad.

Book Recommended for Study

Mutala-e-Ghubar-e-Khatir by Abdul Qavi Dasnavi available from Maktba Jamia, Urdu Bazar, Delhi-6.

PAPER-IV : HISTORY OF URDU LITERATURE, PROSODY AND RHETORIC

Marks : 100
Time : 3 Hrs.

History of Urdu Literature : 60 Marks

Books Recommended for Study :

1. Tarikh-e-Adab Urdu by Ram Babu Saxena.
2. Mulkhtasar Tarikh-e-Urdu by Ejaz Hussain.
3. Aab-e-Hayat by Mohammad Hussain Azad.

Prosody : 20 Marks

The following four meters are to be studied :

1. Mutqarib Salim.
2. Rajab Salim.
3. Hazaj Salim.
4. Ramal Salim.

Rhetoric 20 Marks

1. Tazad.
2. Mar'at-ul-Nazir.
3. Hasan-e-Talil.
4. Tajahua-e-Arifana.
5. Talmih.
6. Laf-O-Nashar.
7. Ishteqaq.
8. Siyaq-ul-Adad.
9. Mubaligha.
10. Tajnis-e-Tam.
11. Tajnis-e-Murakab.
12. Tajnis-e-Zayidi.
13. Tajnis-e-Khatti.
14. Tausif-ul-Sifat.

GERMAN

B.A. (HONOURS) THIRD YEAR EXAMINATION, 2011

PAPER : III

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment (For regular students)	:	10 Marks
Time	:	3 Hrs.

Note :

1. The question paper shall be set in German.
2. The mode of evaluation for Internal Assessment is to be followed as per University Guidelines.

Details of Theory Examination **Max. Marks : 90**

Four out of **six questions** are to be attempted on the History of Literature.

The following movements in Literature are to be studied :

1. *Aufkläung.*
2. *Sturm und Drang.*
3. *Klassik.*
4. *Romantik.*

Courses of Reading :

Prescribed Text Books :

1. Geschichte der deutschen Dichtung by Fricke/Klotz.
2. Wege der deutschen Literatur (Eine geschichtliche Darstellung) by Hermann Glaser, Jakob Lehmann, Arno Lubos.

Reference Book for Supplementary Reading :

Deutsche Literatur geschichte (Von Anfangen bis zur Gegenwart), J.B. Metzler Verlag Stuttgart.

PAPER IV : Viva-Voce Max. Marks : 100

Note : Questions to be asked on topic covered in the prescribed text books.

Details :

1.	Unseen text reading and comprehension	:	40 Marks
2.	To speak on one of the three given topics on the structure of present day German Society.	:	30 Marks
3.	General Conversation.	:	30 Marks
		Total :	**100 Marks**

DEFENCE & STRATEGIC STUDIES

B.A. (HONOURS) THIRD YEAR EXAMINATION, 2011

Outlines of Tests, Syllabi and Courses of Reading

Note : Each theory paper will have one compulsory short answer type question containing 15 questions of 2 marks each covering the entire syllabus. The candidates will be required to attempt any 10 questions. In addition, there will be four sections of the question paper. The candidate will be required to attempt one question from each of these sections carrying 17 ½ marks each.

PAPER-III : AREA STUDIES—CHINA

Max. Marks	:	100
Theory	:	90 marks
Assessment	:	10 marks
Time	:	3 Hrs.

SECTION-I

1. Military Geography of China : Location, size and strategic importance.
2. Economic and Industrial resources (in brief).
3. Population, composition and characteristics of major ethnic groups.

SECTION-II

1. History of China in brief from the CH'IN to Manchus Dynasty.
2. Nationalist Revolution in China under Dr. Sun Yet Sen : Events and effects.

SECTION-III

1. Rise of Chiang Kai Shek.
2. The Nanking Government of China, long march in China.
3. The Civil War, in brief.

SECTION-IV

1. Party-Army Relationship.
2. Communist party and its relationship with the Armed Forces.
3. Higher Defence Organisation.
4. Cultural Revolution.

BOOKS RECOMMENDED :

1. Balance, Edger O. : The Red Army of China, Russell Square, London.
2. Chatterji, B.R. : Modern China –A Short History, Meenakshi Printing, Meerut.
3. Encyclopaedia of Britannica.
4. Gupta, R.S. : History of Modern China, Sterling Pub.
5. Kapur, Harish : China in World Politics, Div. Pub., N.D.
6. Liu, F.F. : A Military History of Modern China, Princeton Uni. Press.
7. Mao Tse Tung : Communist Rule in China.
8. Mc. Aleavy, Henry : The Modern History of China.
9. Mitchison : Chinese Revolution.
10. Nung Chen Li : The Political History of China, Princeton Uni. Press.
11. Pelissier Roger : The Awakening of China, Secker and Warturg, London.
12. Tregear, T.R. : A Geography of China, University of London.
13. Wilson Dick : A Quarter of Mankind, Weidenfeld and Micholson Olson, London.

PAPER-IV : AREA STUDIES—CHINA

Max. Marks	:	100
Theory	:	90 marks
Assessment	:	10 marks
Time	:	3 Hrs.

SECTION-I

SINO-INDIAN WAR (1962)

1. Events Leading to the war.
2. Operations in brief.
3. Lessons learnt.

SECTION-II

1. Determinants of Chinese Foreign Policy.
2. China's Relations with India.
3. China's Relations with Pakistan and its impact on India.

SECTION–III

1. China's Relations with other neighbouring countries.
2. China's Relations with the U.S.A.
3. China's Relations with Russia.

SECTION-IV

1. Defence Potential of China.
2. China's Modernization Program.
3. China's Nuclear Weapon Development.

Books Recommended :

1.	Cassell & Company	:	Guerilla Warfare—Mao–Tse-Tung & Guevera, Red Lion Sq., London.
2.	Dalvi, J.P.	:	Himalyan Blunder, Thacker and Co., Bombay.
3.	Griffith Samoul, B.	:	The Chinese People's Liberation Army, Army Pub., N.D.
4.	Kapur, Harish	:	China in World Politics, Div. Pub., N.D.
5.	Mao Tse Tung	:	Communist Rule in China.
6.	Sharma, B.L.	:	The Pakistan—China Axis, Asia Pub., Bombay.

7.	Sinha, K.K.	:	Problems of Defence of South and East Asia, Rawat Pub., Jaipur.
8.	Trager N. Frank and William Genderson	:	Twentieth Appraisal Year, Weidenfeld and Nicolson, London.

Journals :

1.	Institute of Defence Studies & Analysis (India)	:	(i) Strategic Digest. (ii) Strategic Analysis.
2.		:	The China Quarterly.
3.	The Institute of Strategic Studies	:	Military Balance, IISS, London.
4.	RAUSEN DATTA	:	Problem of Communism.

HISTORY

B.A. (HONOURS) THIRD YEAR EXAMINATION, 2011

Any two papers from the following :

(i) Social and Cultural Trends in Modern India.

(ii) Political Developments in U.S.A. (1775-1865).

(iii) Political Developments in Russia (1801-1933).

(i) SOCIAL AND CULTURAL TRENDS IN MODERN INDIA

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hrs.

GENERAL INSTRUCTIONS :

1. In all, ***nine*** questions will be set. Each question will carry 18 marks.
2. First question shall be Short Answer type containing 15 short questions spread over the whole syllabus. Candidates will attempt ***nine*** out of the fifteen questions in about 25-30 words each. It shall carry 18 marks and shall be **compulsory.**
3. Rest of the paper shall contain 4 units. Each unit shall have two essay type questions and the candidate shall attempt one question from each unit – 4 in all.
4. Each essay type question will be set on half of the topics and not on a single sub-topic.

UNIT-I

1. *Socio-Cultural Trends in Early 19th Century :* Religion, education and literature.
2. *Women Education :* Aligarh Girls School; Kanya Maha Vidayalya.

UNIT-II

3. *Socio-Religious Reform Movements in Punjab and North–West :* The Arya Samaj; Singh Sabhas and the Ahmadiyas.
4. *Socio-Religious Awakening in the Central Belt and Maharashtra :* The Manav Dharma Sabha; the Prarthana Samaj; the Rahnumai Mazdayesan Sabha (among Parsis).

UNIT-III

5. *Cultural Reorientation in South :* Annie Besant and Theosophical Movement.
6. *The Marriage Reform Efforts :* Legislative measures for the abolition of Sati and Infanticide; contribution of reformers towards the promotion of widow-remarriage movement.

UNIT-IV

7. *The Social Protest Movements :* Role of Jyotiba Phule in Maharashtra; Swami Narayana Guru in Kerala; Mangoo Ram and Ad Dharm Movement in Punjab.
8. *The Literary Trends :* Development of literature in Hindi; Bengali and Marathi languages.

Recommended Readings :

1. Kenneth W., Jones : *The New Cambridge History of India : Socio-Religious Reform Movements in British India* (Cambridge, 1989).
2. R.C. Majumdar and Others (ed.) : *British Paramountcy and Indian Renaissance,* Part II, Bhartiya Vidya Bhavan's Series, Vol. X.
3. Robert D. Baird (ed.) : *Religion in Modern India,* Manohar, New Delhi, 1981.
4. Gail Minault : *Secluded Scholars : Women's Education and Muslim Social Reform in Colonial India,* New Delhi, O.U.P., 1998.
5. Geraldine Forbes : *Women in Modern India,* Foundation Books, New Delhi, 1996.
6. Tara Chand : *History of the Freedom Movement in India,* Vol. II, New Delhi, 1967, Publication Division, G.O.I.
7. Mark Jurgensmeyer : *Religious Rebels in the Punjab,* Delhi, Ajanta, 1988.

(ii) POLITICAL DEVELOPMENTS IN U.S.A. (1775-1865)

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hrs.

GENERAL INSTRUCTIONS :

1. In all, ***nine*** questions will be set. Each question will carry 18 marks.
2. First question shall be Short Answer type containing 15 short questions spread over the whole syllabus. Candidates will attempt ***nine*** out of the fifteen questions in about 25 to 30 words each. It shall carry 18 marks and shall be **compulsory.**
3. Rest of the paper shall contain 4 units. Each unit shall have two essay type questions and the candidates shall attempt one question from each unit – 4 in all.
4. Each essay type question will be set on half of the topics and not on a single sub-topic.

UNIT-I

1. *The War of Independence :* Causes; Consequences; The Peace Treaty.
2. *The Making of the Constitution :* Federal Conventions; The Framework of the Constitution and its ratification.

UNIT-II

3. *The New Nation :* Administration of George Washington, Thomas Jefferson; The War of 1812.
4. *Emergent Nationalism :* Foreign Policy of James Monroe.

UNIT-III

5. *Westward Movement :* Nature and results of the Westward Movement.
6. *Expansionism in American Politics :* Manifest Destiny; Causes and results of American Mexican War.

UNIT-IV

7. *The Growth of Social Idealism :* Education, Science and Technology.
8. *Civil War :* Causes; Role of Abraham Lincoln in the Civil War.

Recommended Readings :

1. Henry Bamford Parkes : *The United States of America : A History*, 1964 (Reprint), Scientific Book Agency, Calcutta, 1976.
2. Richard Current & Others : *History of United States : A Survey,* Alfred A. Knopf, New York, 1965.
3. Julius W. Pratt : *A History of United States Foreign Policy,* Prentice Hall (Second Edition), New Jersey, 1965.

(iii) **POLITICAL DEVELOPMENTS IN RUSSIA (1801-1933)**

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hrs.

GENERAL INSTRUCTIONS :

1. In all, ***nine*** questions will be set. Each question will carry 18 marks.
2. First question shall be Short Answer type containing 15 short questions spread over the whole syllabus. Candidates will attempt ***nine*** out of the fifteen questions in about 25 to 30 words each. It shall carry 18 marks and shall be **compulsory.**
3. Rest of the paper shall contain 4 units. Each unit shall have two essay type questions and the candidate shall attempt one question from each unit – 4 in all.
4. Each essay type question will be set on half of the topics and not on a single sub topic.

UNIT-I

1. *Liberalism and Reaction :* Alexander I; Nicholas I.
2. *Modernisation of Russia :* Alexander II; Alexander III & Nicholas II.

UNIT-II

3. *Russia and the Far East :* Interventions in China, Mongolia and Manchuria.
4. *Background of Russian Revolution 1905-1917 :* Revolution of 1905; Russia on the eve of the Revolution of 1917 – Polity, Society, Economy; Lenin's Contribution to the Revolution.

UNIT-III

5. *Causes of the Russian Revolution and its Aftermath :* The February Revolution and the Provisional Government, the Bolshevik (October Revolution), its impact on Russia and the World.
6. *Civil War :* Russian withdrawal from First World War; Peace of Brest Litovisk; Allied Intervention.

UNIT-IV

7. *The Soviet Economy :* War Communism; New Economic Policy and its impact.
8. *Planned Economy :* Weaning from the New Economic Policy and the First Five Year Plan.

Recommended Readings :

1.	Washington Moore Junior	:	*The Social Origins of Dictatorship and Democracy.*
2.	Paul Kannedy	:	*The Rise & Fall of Great Powers,* Fontana Press London, 1988.
3.	Lionel Kochan	:	*The Making of Modern Russia,* Penguin Books, 1962.
4.	Basil, Dymtrystyn	:	*History of Russia,* Prentice Hall, New Delhi, 1977.

Reference Readings :

1.	E.H. Carr	:	*The Bolshevik Revolution,* Penguin Books Ltd., Victoria, 1971 (Reprint).
2.	M.T. Florinsky	:	*Russia : A Short History,* Macmillan, London, 1971 (2nd ed.)

POLITICAL SCIENCE

B.A. (HONOURS) THIRD YEAR EXAMINATION, 2011

Outlines of Tests, Syllabi and Courses of Reading

PAPER III	:	Option (i)	International Relations since 1945.
		Option (ii)	Modern Indian Political Thought.
		Option (iii)	Women and Political Process.
PAPER IV	:	Option (i)	Research Methods in (Political Science) Social Sciences.
		Option (ii)	Modern Political Ideologies.
		Option (iii)	Western Political Thought.

PAPER III : Option (i) — INTERNATIONAL RELATIONS SINCE 1945

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hrs.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

The syllabus has been divided into four units.

There shall be 9 questions in all. The first question shall be short answer type containing 15 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions i.e. 2 marks each. It shall carry 18 marks and shall be a **compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all, of 18 marks each.

Objectives : This paper aims at mapping out some of the major transformations that have occurred in the realm of International Relations since 1945.

UNIT-I Post World War-II International Relations :

(a) Post War World Order.

(b) Emergence of Super Powers.

UNIT-II Democracy and Rise of IIIrd World :

(a) Non-Alignment : Meaning and Relevance.

(b) Decolonization and Afro Asian Regimes.

UNIT-III Transformation in Post World War II International Relations :

(a) Decline of Bipolarity.

(b) Rise of Neo-colonialism.

UNIT-IV New Global Scenario and International Relations :

(a) Demand for New World Order.

(b) Critique of Global Terrorism and Democracy.

Suggested Readings :

1. Peter Calvocoressi : *World Politics Since 1945*, London : Longman, 1945.
2. Budhraj, V.S. : *International Relations After Second World War* (in Hindi), Chandigarh, Haryana Sahitya Academy, 2000.
3. Keswani, K. B. : *International Relations in Modern World* (1900-1988), Bombay, Himalaya Publishing House, 1994.
4. Baylis and Steve Smith : *The Globalization of World Politics – An Introduction to International Relations*, Oxford, Oxford University Press, 2001.
5. Alan C. Lamborn and Joseph Lepgold : *World Politics into the Twenty First Century : Unique Context, Enduring Patterns,* New Jersey, Prentice Hall, 2003.

References :

1. Paul Kennedy : *The Rise and Fall of Great Power*, Fontana Press.
2. Henry Kissinger : *Diplomacy*, London, Simon & Soluster, 1994.

PAPER III : Option (ii) — MODERN INDIAN POLITICAL THOUGHT

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hrs.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

The syllabus has been divided into four units.

There shall be 9 questions in all. The first question shall be short answer type containing 15 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions i.e. 2 marks each. It shall carry 18 marks and shall be a **compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all, of 18 marks each.

Objectives : The paper aims at creating understanding about the evolution of political thought in modern India with special reference to the stated philosophers.

UNIT-I Sir Syed Ahmed Khan :

(a) Readings and Interpretations of 1857.

(b) Education.

(c) Muslim Nationalism.

UNIT-II M.K. Gandhi :

(a) Satyagraha.

(b) Swaraj.

(c) Trusteeship and Non-Violence.

UNIT-III M.N. Roy :

(a) Critique and Contribution to Communism.

(b) Radical Humanism.

UNIT-IV B.R. Ambedkar :

(a) Annihilation of Caste.

(b) Social Justice.

(c) Democracy : Political and Social.

Recommended Readings :

1. Thomas Pantham & Deutsch (ed.) : *Political Thought in Modern India*, New Delhi, Sage, 1986.

2. Majumdar, B.B. : *History of Indian Social and Political Ideas*, Calcutta : Bookland, 1967.

References :

1. Parekh, Bhikhu : *Colonialism, Tradition, Reform : Analysis of Gandhi's Political Discourse*, New Delhi, Sage, 1989.

2. Iyer, Raghvan (ed.) : *Moral and Political Thought of Mahatma Gandhi*, New York, Oxford, 1973.

3. Appadorai, A. : *Indian Political Thinkers in the Twentieth Century,* Delhi, Oxford, 1972.

4. Jatava, D.R. : *The Political Philosophy of B.R. Ambedkar,* Delhi; Arya, Phoenix, 1965.

PAPER III : Option (iii) — WOMEN AND POLITICAL PROCESS

Max. Marks		: 100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hrs.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

The syllabus has been divided into four units.

There shall be 9 questions in all. The first question shall be short answer type containing 15 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions i.e. 2 marks each. It shall carry 18 marks and shall be a **compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all, of 18 marks each.

Objectives : The paper aims to introduce students to the key approaches of Feminist thought contemporary debates concerning women and politics in India. It will include an analysis of the Indian State, the role of women's movement and the major feminist issues in contemporary Indian politics.

UNIT-I **Theoretical Approaches to Women Studies :**

(a) Liberal Approach.

(b) Radical Feminism - Critique of Patriarchy.

(c) Marxist Approach : Socialist Feminism .

UNIT-II **Rise of Women's Movement, its role in Contemporary Societies—Western, Socialist and Third World with special reference to India :**

Historical Legacy :

(a) Women's Movement at the Global level; Nature, Issues and Contemporary debates.

(b) National Liberation Movement: With special reference to contribution of women in India.

UNIT-III Women and Society in India :

(a) Evolution of Women's position in India – a historical overview.

(b) Impact of Development Process on Women.

(c) Social Position :

(i) Demographic pattern.

(ii) Legal Political and Property Rights.

(iii) Access to Education.

UNIT-IV Women's Movement in Contemporary India :

(a) Women's Organisations–Radical, Socialist and Others.

(b) Issue of Autonomous Women's Movement.

Recommended Readings :

1. Millet, Kate, *Sexual Politics,* New York, Doubleday, 1970.

2. Michele Barrett, *Women's Oppression Today,* London, Verso, 1980.

3. Lenin, V.I., *On the Emancipation of Women*, Moscow, Progress, 1981.

4. Eisenstein, Zillah, *Capitalist Patriarchy and the Case for Socialist Feminism,* New York, Longman, 1981 (also New York, Monthly Review Press, 1979).

5. Boserup, Easter, *Women's Role in Economic Development,* London, George Allen & Unwin, 1975.

6. Altback, E.M. (Ed.), *From Feminism to Liberation*, Cambridge, Massachusetts, Schenkman Hare Revised Ed., 1980.

7. Jayawardhane, Kumari (Ed.), *Feminism and Nationalism in the Third World,* New Delhi, 1987.

8. Phadnis, Urmilla and Malani Indira eds., *Women of the World : Illusion and Reality,* New Delhi, Vikas, 1978.

9. Desai, N., *Women in Modern India,* Bombay, Vora & Co., 1977.

10. Nanda, B.R. ed., *Indian Women from Purdah to Modernity,* New Delhi, Vikas, 1976.

11. *I.C.S.S.R., Status of Women's Committee Report,* New Delhi, ICSSR, 1975.

12. Mazumdar, Vina, *Towards Equality,* ICSSR, Delhi.

13. Desai, Neera and Krishanaraj, M., *Women and Society in India,* Delhi, Ajanta, 1987.

14. Chakaravarty, Shanti, *Rural Women's Claim to Priority,* New Delhi, C.W.D.S., 1986.

15. "Women's Struggles and Movement", Paper of Third National Conference, Chandigarh, IAWS.

16. Kaushik, S. (ed.), *Women's Oppression, Patterns and Perspective,* Vikas, New Delhi, 1984.

17. Kumar, Radha, *A History of Doing,* Kali Publications, New Delhi, 2006.

18. Gandhi, Nandita & Shah, Nandita, *The Issues at State*, New Delhi, Kali Publication, 1992.

19. Menon, Nivedita (ed.), *Gender & Politics in India*, New Delhi, Oxford University Press, 1999.

20. Rao, Anupama (ed.), *Gender & Caste*, Delhi, Sage Publications, 2005.

PAPER IV : Option (i) — RESEARCH METHODS IN (POLITICAL SCIENCE) SOCIAL SCIENCES

Max. Marks		: 100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hrs.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

The syllabus has been divided into four units.

There shall be 9 questions in all. The first question shall be short answer type containing 15 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions i.e. 2 marks each. It shall carry 18 marks and shall be a **compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all, of 18 marks each.

Objectives : The paper endeavours to introduce the students about the tools of research methodology into social sciences. The paper would enable the students to take up research after Master's degree.

UNIT-I Research in Social Science with reference to Political Science—Definition, Nature and Limitations.

UNIT-II Methods – Traditional and Scientific, Survey Research Methods : Observation, Interviews, Questionnaire, Content–Analysis.

UNIT-III (a) Sample and Sampling Techniques.

(b) Measures of Central Tendency—Mean, Mode and Median.

(c) Data Processing—Coding Tabulation and Analysis.

UNIT-IV Research Design : Formulation of Research problem, Hypotheses, Objective, relevance of study, Literature Review, Methodology Report Writing.

Suggested Readings :

1. Goode and Hatt, *Methods in Social Research,* Auddond (McGraw - Hill), New York, 1952.

2. Pauline Yong, *Scientific Survey and Research,* New Delhi, 1968.

3. Arnold Brecht, *Political Theory,* Times of India Press, 1970.

4. Michael S. Lewis-Beck, ed., *The Encyclopedia of Social Science Research Methods,* New Delhi, Sage, 2004.

5. Keith F. Punch, *Introduction to Social Research,* London, Sage, 2005.

PAPER IV : Option (ii) : MODERN POLITICAL IDEOLOGIES

Max. Marks		: 100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hrs.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

The syllabus has been divided into four units.
There shall be 9 questions in all. The first question shall be short answer type containing 15 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions i.e. 2 marks each. It shall carry 18 marks and shall be a **compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all, of 18 marks each.

Objectives : The paper refers to the basic tenets of modern political ideologies.

UNIT-I **Liberalism :**

Classical and Modern Liberalism, New Problems of Contemporary Liberalism.

UNIT-II **Socialism :**

Marxian Socialism.

Fabianism.

Syndicalism and Guild Socialism.

Democratic Socialism.

UNIT-III Anarchism.

Fascism.

Modern Conservatism.

UNIT-IV **Gandhism :**

Social and Political Ideas of Gandhi and his successors.

Suggested Readings :

1. Bandhyopadhya, J., *Political and Social Thought of Gandhi,* Bombay, 1969.
2. George Lichtheim, *Marxism : An Historical and Critical Study,* Routledge and Kegan Paul, London, 1962.
3. Noel O' Sullivan, *Conservation*, St. Martin's Press, New York, 1976.
4. Manning, D.J., *Liberalism*, St. Martin's Press, New York, 1976.

PAPER IV : Option (iii) — WESTERN POLITICAL THOUGHT

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hrs.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

The syllabus has been divided into four units.
There shall be 9 questions in all. The first question shall be short answer type containing 15 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions i.e. 2 marks each. It shall carry 18 marks and shall be a **compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all, of 18 marks each.

Objectives : This paper aims to introduce to the students the major themes of Western Political Thought. This will be done by taking an in-depth study of the key thinkers of this tradition.

UNIT-I Hobbes.

Nature of Man, State of Nature, Natural Laws.

Social Contract.

Basis of Political Obligation.

UNIT-II Locke.

Nature of Man, State of Nature, Natural Rights.

Social Contract.

Contribution of Locke to Liberalism.

UNIT-III Rousseau.

State of Nature, Natural Man.

Social Contract.

Theory of General Will.

UNIT-IV Karl Marx.

Historical Materialism, Economic Determinism.

Theory of Surplus Value.

Theory of Revolution.

Recommended Readings :

1. Sabine, G.H., *A History of Political Theory,* Bombay : Oxford & IBH, 1973.

2. Dunning, W.A., *A History of Political Theories,* 3 Vols., Allahabad : Central, 1973.

3. Sait, E.M. (ed.), *Masters of Political Thought,* 3 Vols., London : George and Harap, 1947.

4. William Ebenstien, *Great Political Thinkers—Plato to the Present,* OUP : New Delhi, 1966.

5. Jen Pike, Nigel Waaburton & Derek Matravers, *Reading Political Philosophy, Machiavelli to Mill*, London, Routledge, Open University, 2000.

6. Andrew Hacker, *Political Theory*, London, OUP, 1961.

7. David Boucher & Paul Kelly, *Political Thinkers, From Socrates to the Present*, Paperback U.K., 2003.

8. David Boucher & Paul Kelly, *The Social Contract—from Hobbes to Rawls*, Paperback, U.K., 1994.

References :

1. Rapheal, D.D., Hobbes, *Morals and Politics,* London, George Allen, 1978.

2. David Macdellan, *Marx,* London, Fontana Modern Masters Series, 1983.

3. Plamenatz, J., *Man and Society,* 2 Vols., New York, Longman, 1984.

4. Hall, J.C., *Rousseau : An Introduction to his Political Philosophy,* London, MacMillan, 1973.

5. John, Dunn, *The Political Thought of John Locke,* Cambridge, Cambridge University Press, 1969.

6. Roger, D. Masters, *The Political Philosophy of Rousseau,* Princeton, Princeton University Press, 1968.

7. Shlomo, Avineri, *The Social and Political Thought of Karl Marx,* New Delhi, S. Chand, 1968.

8. Jon Elster, *Making Sense of Marx,* Cambridge, Cambridge University Press, 1985.

9. Ernest Cassirer, *The Question of Jean Jacques Rousseau,* New Haven and London, Yale University Press, 1989.

ECONOMICS

B.A. (HONOURS) THIRD YEAR EXAMINATION, 2011

The candidates are required to study **two** papers out of the following five papers :

I Agricultural Marketing.

II Money and Banking.

III International Economics.

IV Public Finance.

V Introduction to Econometrics.

Paper–I : AGRICULTURAL MARKETING

Max. Marks		: 100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours
Total Teaching Hours	:	60

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

The syllabus has been divided into four units.

There shall be 9 questions in all. All questions carry equal marks. First question shall be short answer type containing 12 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions i.e. 2 marks of each. It shall carry 18 marks and shall be **compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall *attempt one question* from each Unit – 4 in all.

UNIT-I

Marketing : Definitions, scope, marketed and marketable surplus, factors affecting marketable surplus, size, distribution of marketable surplus, nature of demand and supply of agricultural commodities.

UNIT-II

Mobilization of agricultural surpluses, terms of trade between agriculture and industry, income and price elasticities of agricultural commodities, Cobb-Web model.

UNIT-III

Structure and functioning of agricultural markets, marketing margin, marketing efficiency, measures to improve marketing efficiency.

Foodgrain marketing system in India, State intervention in foodgrains marketing, role of different agencies

(FCI, State agencies, State department).

UNIT-IV

Marketing of major agricultural commodities in India (foodgrains, commercial crops). Price expectation and price uncertainties, price stabilization measures. Finance of agricultural marketing.

Recommended Readings :

1. A.S. Kahlon & M. V. George : *Agricultural Marketing and Price Policies,* Allied Publishers, New Delhi, 1985.
2. A.S. Kahlon & D. S. Tyagi : *Agricultural Price Policy in India,* Allied Publishers, New Delhi, 1983.
3. Dala C. Dahl & Jerema W. Hummod : *Market and Price Analysis – The Agricultural Industries,* McGraw Hills Book Company, 1977.

Supplementary Readings :

1. Richard, Nelson : *Marketing of Agricultural Production,* Macmillan, New York, 1968.
2. Rajbans Kaur : *Agricultural Pricing Policies in Developing Countries,* Kalyani Publishers, Ludhiana, 1984.
3. Uma J. Lele : *Foodgrain Marketing in India,* Popular Prakashan, Bombay, 1973.

Paper-II : MONEY AND BANKING

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours
Total Teaching Hours	:	60

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

The syllabus has been divided into four units.

There shall be 9 questions in all. All questions carry equal marks. First question shall be short answer type containing 12 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions i.e. 2 marks of each. It shall carry 18 marks and shall be **compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall *attempt one question* from each Unit – 4 in all.

UNIT-I

Money and near money. Demand for money; quantity and cash balance theories of money, latest development in the theory of money, Friedman, and Gurley and Shaw approaches.

UNIT-II

Supply of Money : Components & determinants, money multiplier, structure of rate of interest approaches. IS and LM functions. Neutrality and non-neutrality of money. Integration of theory of value and theory of money.

UNIT-III

Banking : Credit creation; commercial banks; their functions and importance. Central banking; functions and techniques of monetary management.

Financing institutions and their role. Role of non-banking financial intermediaries. Monetary policy in India—Objectives and instruments.

UNIT-IV

International Monetary Cooperation, International Monetary Fund. International Bank for Reconstruction and Development. International Finance Corporation. International Development Association. Asian Development Bank and the International Institutions.

Suggested Readings :

1. Luckett, D. C. : *Money and Banking.*
2. Thorn, R. S. : *Introduction to Money and Banking.*
3. Laidler, D. E. W. : *The Demand for Money : Theories and Evidence,* Allied Publishers, New Delhi, 1977.
4. Gupta, S. B. : *Monetary Planning in India,* S. Chand, New Delhi, 2004.

Paper-III : INTERNATIONAL ECONOMICS

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours
Total Teaching Hours	:	60

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

The syllabus has been divided into four units.

There shall be 9 questions in all. All questions carry equal marks. First question shall be short answer type containing 12 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions i.e. 2 marks of each. It shall

carry 18 marks and shall be **compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall *attempt one question* from each Unit – 4 in all.

UNIT-I

The Pure Theory of International Trade : Theories of Absolute Advantage, Comparative Advantage and Opportunity Costs. Heckscher-Ohlin Theory. Factor Price Equalisation. Gains from Trade, their measurement and distribution. Terms of trade, secular : deterioration hypothesis.

UNIT-II

Commercial Policy : The theory of tariffs, optimum and effective rate of tariff, inter-relationship between trade, aid and economic development. The theory of economic integration among nations; forms of economic integration. The theory of Customs Union.

UNIT-III

Balance of Payments, Exchange Rate and Trade Structures : The process of adjustment in the balance of payments. The concept of foreign trade multiplier. Fixed Vs. flexible exchange rates and their relative merits and demerits. Structure of trade, trade between developing and developed countries. Unequal Exchange and its impact on Balance of Payments. Current problems of Trade and Finance of developing countries. Role of multinational corporations in the developing countries.

UNIT-IV

International Monetary System : IMF and international monetary system, present international monetary system, problems of international liquidity. Proposals for international monetary reforms. New International Economic Order.

Recommended Readings :

1. B. O. Soderston : International Economics, 2nd Edition, Macmillan Press, London, Reprint, 1994.
2. Ranjit Singh Ghuman : Antar-Rashtriya Arth Vigiyan, (Punjabi Medium) Punjabi University, Patiala.

Supplementary Readings :

1. C. P. Kindleberger : International Economics, Richard Irwin, Homeswood, Illinois, Indian Edition, 1987.

2. Sir Rai Harrod (Trans. Pritam Singh Gill) : Antar-Rashtriya Arth Shastar, Punjabi University, Patiala.

Paper-IV : PUBLIC FINANCE

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours
Total Teaching Hours	:	60

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

The syllabus has been divided into four units.

There shall be 9 questions in all. All questions carry equal marks. First question shall be short answer type containing 12 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions i.e. 2 marks of each. It shall carry 18 marks and shall be **compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall *attempt one question* from each Unit – 4 in all.

UNIT-I

The Principle of Public Finance – Subjective and Objective criteria.

Theory of Public Expenditure : Maximum Social Advantage, Theory of Social Goods : Distinction between public, private and merit goods. Public expenditure and economic growth. Effects of public expenditure on production, distribution, social over-head capital, stability and Innovation. Criteria of Public Expenditure.

UNIT-II

Theory of Public Revenue : Theories of Taxation : Benefits principle, cost of service principle, ability to pay theory, principle of equity. Effects of taxation. Analysis of major taxes – Income Tax, Expenditure Tax, Corporation Tax, Wealth Tax, Excise Duties, Sales Tax, and Custom Duties. The concept of taxable capacity. Taxation in a developing economy. Fiscal incidence : Theories of tax shifting, concepts of incidence, measurement of incidence.

UNIT-III

Public Debt : Role of Public Debt. Different forms of Public Debt. Burden of Public Debt, Deduction of Public Debt. Debt management.

Budgetary Policy : Functional and Economic Classification of budgets and their uses. Performance budgets. Budget as an instrument of mobilization and canalization of resources and redistribution of income and wealth.

UNIT-IV

Balance and unbalanced budgets.

Fiscal Federalism : Principles of federal finance.

Development Finance : Functional Finance vs. Development Finance.

Mobilisation of financial resources for planned development.

Suggested Readings :

1. Musgrave, R. A. : *Theory of Public Finance.*
2. Taylor & Phillip : *The Economics of Public Finance.*
3. Buchanan, J. M. : *The Public Finance.*

Paper-V : INTRODUCTION TO ECONOMETRICS

Max. Marks		: 100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours
Total Teaching Hours	:	60

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

The syllabus has been divided into four units.

There shall be 9 questions in all. All questions carry equal marks. First question shall be short answer type containing 12 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions i.e. 2 marks of each. It shall

carry 18 marks and shall be **compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall *attempt one question* from each Unit – 4 in all.

UNIT-I

Definition and Scope of econometrics. The methodology of economic research. Specification and estimation of an econometric model. Random variable and probability distribution. Basic concepts : unbiasedness, efficiency and consistency.

Single Equation Models : Ordinary Least Squares Estimation.

UNIT-II

Testing of Hypothesis : Basic Concepts; Type I and Type II errors. Simple and composite hypothesis. t and F tests. Maximum Likelihood estimation.

UNIT-III

The General Linear Regression model in Matrix form. Estimation and properties of the estimators. Tests of significance of coefficients. Partial Correlation Coefficients.

Multicollinearity : Its detection and solution.

Heteroscedasticity : Its detection and solution.

UNIT-IV

Dummy Variables : The dummy variable trap. Interpretation of slope and intercept.

Lagged Variables : Uses in economics. Estimation Problems, Koyck transformation, Partial and Stock adjustment models.

Estimation of demand function.

Estimation of consumption function.

Recommended Readings :

1. Johnson, J. : *Econometrics Methods,* 2nd Edition, McGraw Hill, New York, 1984.

2. Gujarati, Damodar : *Basic Econometrics,* McGraw Hill, New Delhi, 2003.

Supplementary Readings :

1. Kendall, M. G. & A. Sturat (eds.) : *Advanced Theory of Statistics,* Vols. I & II, Griffin and Co., London.

2. Intrilligator, M. D. : *Econometric Models and Applications,* Prentice Hall.

3. Maddala, G. S. : *Econometrics,* New Delhi, McGraw Hill, 1977.

4. Wallis, K. F. : *Topics in Applied Econometrics,* London, Bray Hills.

SOCIOLOGY

B.A. (HONOURS) THIRD YEAR EXAMINATION, 2011

Paper III (SOC 351) : Methods and Techniques of Social Research (Compulsory)

Paper IV : Any **one** from the following two options :

Option (i) (SOC 352) Social Demography (Written : 90 marks)
(Internal assessment :10 marks)

Option (ii) (SOC 353) Social Deviance (Written : 90 marks)
(Internal assessment :10 marks)

PAPER III : METHODS AND TECHNIQUES OF SOCIAL RESEARCH

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hrs.

Note :

1. For written paper the students will be required to attempt five questions in all. Question No. I will be compulsory and cover the entire syllabus. In addition, students will attempt **four** essay type questions, one from each of the four units.

2. Question No. I will be compulsory comprising of short answer questions. It will consist of 12 short answer type questions of 2 marks each, out of which the students are required to attempt 9 short answer type questions ($9 \times 2 = 18$ marks).

3. On an average, 15 hours are to be devoted to each unit.

4. Question Nos. II to IX will be long answer questions, two questions from each unit with internal choice, consisting of 18 marks each ($4 \times 18 = 72$ marks).

Objective :

The students will be introduced with basic methods and techniques in social research. The objective of this course is to familiarize the students with the basic elements of scientific method, various methods of data collection and social measurement.

Course Content :

UNIT-I

Scientific Method : Features; Concepts; Stages of social research; Hypothesis.

UNIT-II

Sampling Techniques: Definition and Concepts (population/stratum/census) Sampling – Types: Probability: Simple random/stratified random/cluster/multi-stage. Non–Probability: Accidental/Quota/Purposive; Sampling Error.

UNIT-III

Techniques of Data Collection : Observation, Participant and Non-Participant Observation, Interview Schedule and Questionnaire (Structured/Unstructured); Case study method.

UNIT-IV

Measurement : Levels and Problems, Measures of Central Tendency: Mean, Mode, Median.

Essential Readings :

1. Sellltiz, Claire et.al : *Research Methods in Social Relations* (Revised ed.), New York: Henry Hold & Co. (latest edition).
2. Goode, W.J. & Hatt, P.K. : *Methods in Social Research,* New York: McGraw Hill, Latest Indian Edition, 1952.
3. Young, P.V. : Sc*ientific Social Surveys and Research,* New Delhi: Prentice Hall of India Private Ltd., 1988.
4. Mukerjee, P.N. : *Methodology on Social Research: Dilemmas and Perspectives,* New Delhi: Sage, 2000.

Further Readings :

1. Festinger, L. & Katz, D. : *Research Methods in Behavioural Science,* New York: Dryden Press, 1953.
2. Lindzey, Gardner & Aronson : *The Handbook of Social Psychology* (2nd ed.), Addison: Wesley Pub. Comp., 1968.
3. Robson, Colin : *Real World Research*, Oxford: Blackwell, 2000.

PAPER-IV: Option (i): SOCIAL DEMOGRAPHY

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hrs.

Note :

1. For written paper the students will be required to attempt five questions in all. Question No. I will be compulsory and cover the entire syllabus. In addition, students will attempt **four** essay type questions, one from each of the four units.

2. Question No. I will be compulsory comprising of short answer questions. It will consist of 12 short answer type questions of 2 marks each, out of which the students are required to attempt 9 short answer type questions ($9 \times 2 = 18$ marks).

3. On an average, 15 hours are to be devoted to each unit.

4. Question Nos. II to IX will be long answer questions, two questions from each unit with internal choice, consisting of 18 marks each ($4 \times 18 = 72$ marks).

Objective :

The course exposes students to various components of population growth, data sources and basic theoretical problems of population. Concepts of mortality and fertility are discussed in general and with reference to India in particular. Students are apprised of the latest Population Policy in India.

Course Content :

UNIT-I

Population : Definition, Scope, Nature, Sources of Population Data-Census and Vital Registration.

UNIT-II

Population Theories : Malthusian theory of Population; Biological: Theory of Density Principle and Diet Theory; Social Theories: Social Capillarity and Theory of Diffusion: Theory of Demographic Transition.

UNIT-III

Concepts of Mortality : Causes of Mortality, Infant Mortality, Mortality in India.

Fertility : Concept of Fertility, Factors affecting Fertility–Biological, Socio-Cultural and Psychological Fertility in India.

UNIT-IV

Population Policy in India, Female Foeticide (reference to PNDT Act).

Essential Readings :

1. Bhende, Asha & Kanitkar, Tara : *Principles of Population* (revised ed.), Bombay: Tara Himalaya Publishing House, 1996.
2. Bogue, Donald, J. : *Principles of Demography,* New York: John Willey, 1969.
3. Nam, Charles, P. : *Population and Society, 1968.*
4. Premi, M.K. : *An Introduction to Social Demography,* Delhi: Vikas Publishing House, 1983.
5. Weeks, J.R. : *Population: An Introduction to Concepts & Issues,* London: Wardworth Publishing Co., 1988.

Further Readings :

1. Bose, Ashish : *India & The Asian Perspective,* New Delhi: B.R. Publishing Corporation, 1993.
2. Matras, Juda : *A Sociological Approach,* New Jersey: Prentice Hall, 1977.
3. Government of India : *National Population Policy* (2000), Ministry of Health & Family Welfare, GOI.

   (Latest Census Reports and Sample Registration Bulletins of the Government of India.)
4. Premi, M.K. : *India's Population: Heading towards a Billion,* Delhi: B.R. Publications, 1991.

PAPER IV: Option (ii): SOCIAL DEVIANCE

Max. Marks		: 100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hrs.

Note :

1. For written paper the students will be required to attempt five questions in all. Question No. I will be compulsory and cover the entire syllabus. In addition, students will attempt **four** essay type questions, one from each of the four units.

2. Question No. I will be compulsory comprising of short answer questions. It will consist of 12 short answer type questions of 2 marks each, out of which the students are required to attempt 9 short answer type questions ($9 \times 2 = 18$ marks).

3. On an average, 15 hours are to be devoted to each unit.

4. Question Nos. II to IX will be long answer questions, two questions from each unit with internal choice, consisting of 18 marks each ($4 \times 18 = 72$ marks).

Objective :

To understand the contemporary social problems in society, the students are exposed to the basic concept of deviance, its theoretical perspectives and control mechanisms.

Course Content :

UNIT-I

Deviance : Definition, Concept (Relative and Absolute) Distinction between deviant behaviour and deviant role. Pathological theories (Biological and Psychological).

UNIT-II

Sociological Theories : Theories of social disorganization (C.H. Cooley, W.I. Thomas and Florian Znaniecki, W.F. Ogburn).

Structural-Functional Theories : Theory of Anomie (Emile Durkheim, Robert K. Merton), Differential Association (E. Sutherland and D.R. Cressey).

Interactionist Theories : Theory of Labelling (Edwin M. Lemert and Howard S. Becker).-+

UNIT-III

Controlling Deviance : Socialization - means and agencies, Social Control: means and agencies, Social codes and customs; Sympathy: passive and active.

UNIT-IV

Selected Problems of Deviance: Sex Deviance, Delinquency, White Collar Crimes, Cyber Crime.

Essential Readings :

1. Becker, Howard S. : *Outsiders: Studies in the Sociology of Deviance,* Glenco, Free Press, 1987.
2. Ferrell, R. & V.L. Swigert [ed.] : *Social Deviance*, New York: J.B. Lippincott Co. 1975.
3. Haralambos, M. : *Sociology – Themes & Perspectives,* Delhi, Oxford University Press, 1984.
4. Kelly, Delos H. : *Deviant Behaviour: Readings in Sociology of Deviance*, New York: Martin's Press, 1979.
5. Lawson, T & T. Heaton : C*rime & Deviance,* New Delhi : Macmillan, 1999.
6. Metta, Spencer : *Foundations of Modern Sociology*, New Jersey: Prentice Hall Inc., Englewood Cliff, 1979.
7. Michael Mc Kee and Ian Robertson : *Social Problems,* Toronto: Random House Inc., 1976.
8. Thio, Alex : *Deviant Behaviour,* London : Houghton Mifflin, 1978.

Further Readings :

1. Erdwin. H. Pfuhl : *The Deviance Process,* New York: D.Van Nostrand Co., 1980.
2. Ian Robertson : *Sociology,* New York: Publisher Inc. 1977.

3.	John, J. Grant	:	*Social Problems as Human Concern,* San Francisco: Body & Fraser Publishing Company, 1976.
4.	Rubington, Weinberg	:	*Studies of Social Problems,* New York: Oxford University Press, 1977.
5.	Rushing, William A. [ed.]	:	*Deviant Behavior & Social Process:* Rand McNally College Pub. Co., 1985.

PHILOSOPHY

B.A. (HONOURS) THIRD YEAR EXAMINATION, 2011

Paper III Option (i) : PHILOSOPHY OF SCIENCE

Max. Marks		: 100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hrs.

AIMS AND OBJECTIVES :

The aim of this paper is to analyse the concepts of philosophy and science, Einstein's Theory of Relativity i.e. space, time and relativity; Nature of observational language and theoretical language; causality, prediction and probability; Principle of Complimentarity.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

The syllabus has been divided into four units.

There shall be 9 questions in all. First question shall be short answer type containing 15 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions i.e. 2 marks of each. It shall carry 18 marks and shall be **compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall *attempt one question* from each Unit – 4 in all, of 18 marks each.

Course Contents :

UNIT-I

1. The Nature and Scope of Philosophy of Science. Relation between Philosophy and Science.
2. Einstein's Theory of Relativity : Space, time, reality.
3. Methodological character of theoretical concepts. Observational language, theoretical language, a criterion of significance of theoretical terms. Dispositional concepts, Difference between theoretical terms and pure dispositional terms.

UNIT-II

4. Causality, Prediction and Probability.
5. Principle of Causality.
6. Causality as the existence of Laws.

UNIT-III

7. Causal Law and Statistical Laws.
8. Induction and Probability.

UNIT-IV

9. Theories of Probability.
10. Principle of Complimentarity.

Essential Readings :

1. Bridgman, P.W. : *The Logic of Modern Physics* (New York : Macmillan, 1958).
2. Hans Reichenbac : *Rise of Scientific Philosophy* (California : University Press, 1994).
3. Earnest Nagel : *Structure of Science* (Hackett Pub. Co., 1979).

Further Readings :

4. Minnesota Studies in the Philosophy of Science, Minnesota Centre for Philosophy of Science, 1958.
5. Phillip Frank : *Philosophy of Science : The Link between Science and Philosophy* (Englewood Cliff : Prentice Hall, 1957).

Paper III Option (ii) : SOCIAL AND POLITICAL PHILOSOPHY

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hrs.

AIMS AND OBJECTIVES :

The paper aims at understanding the nature of social and political philosophy and some of its fundamental concepts; its major social and political theories and Indian political thought.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

The syllabus has been divided into four units.

There shall be 9 questions in all. First question shall be short answer type containing 15 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions i.e. 2 marks of each. It shall carry 18 marks and shall be **compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall *attempt one question* from each Unit – 4 in all, of 18 marks each.

Course Contents :

UNIT-I

1. Nature and Scope of Social and Political Philosophy :

   (a) Relation between Social and Political Philosophy and Social Sciences such as Sociology and Political Science with special emphasis on the distinction between Philosophy and Science.

   (b) Empirical and Scientific study of society and politics as distinct from philosophical study of society and politics.

UNIT-II

2. Some Fundamental Concepts of Social and Political Philosophy :

   (i) Distinction between Natural Facts and Social Facts.

   (ii) Normative Aspects of Socio-Political Reality.

   (iii) State and Civil Society.

   (iv) Liberty, Equality and Fraternity.

   (v) Social Change and Social Progress.

   (vi) Tradition and Modernisation.

   (vii) Power and Authority.

   (viii) Reform and Revolution.

   (ix) Justice.

UNIT-III

3. Major Social and Political Theories

   (a) Liberalism.

   (b) Egalitarianism.

   (c) Anarchism.

UNIT-IV

4. Indian Political Thought : Chanakya, Vivekananda, Gandhi, Ambedkar.

Essential Readings :

1. Complete Works of Swami Vivekananda (relevant portions) Advaita Ashram, Calcutta.
2. Tagore : *Religion of Man,* Unwin Books, London.
3. Bhattacharyya, K.C. : *Studies in Philosophy.*
4. Ambedkar, B.R. : *Writings and Speeches,* Government of Maharastra, Bombay.
5. Gautam, S.P. : *Samaj Darshan,* Haryana Sahitya Akademy, Chandigarh.

PAPER IV Option (i) : LOGIC

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hrs.

AIMS AND OBJECTIVES :

The paper aims at training the students in skills of critical reasoning by introducing laws of thought, criteria for validity of arguments, deductive and inductive reasoning.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

The syllabus has been divided into four units.

There shall be 9 questions in all. First question shall be short answer type containing 15 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions i.e. 2 marks of each. It shall carry 18 marks and shall be **compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall *attempt one question* from each Unit – 4 in all, of 18 marks each.

Course Contents :

UNIT-I

1. Nature and Scope of Logic.
2. Deductive Inference.
3. Laws of Thought.

UNIT-II

4. Aristotelian Classification of Propositions.
5. Immediate and Mediate Inference.
6. Syllogism.

UNIT-III

7. Truth Tables.
8. Propositional Calculus.
9. Quantifiers and Propositional Functions.
10. Predicate Calculus.

UNIT-IV

11. Inductive Logic.
12. Probability.
13. Scientific Method.
14. Law and Explanation.

Essential Readings :

1. Irving Copi : *Introduction to Logic,* Prentice Hall, 2004.
2. Nagel and Cohen : *Introduction to Logic and Scientific Method,* Simon Publications, 2002.
3. Irving Copi : *Symbolic Logic,* Prentice Hall, 1979.

OR

Paper-IV Option (ii) : PHILOSOPHY OF CULTURE

Max. Marks		: 100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hrs.

AIMS AND OBJECTIVES :

This paper aims to introduce the problems like origin and meaning of culture, civilization, tradition, the role of ideology and Utopia, Indian cultural tradition.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

The syllabus has been divided into four units.

There shall be 9 questions in all. First question shall be short answer type containing 15 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions i.e. 2 marks of each. It shall carry 18 marks and shall be **compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall *attempt one question* from each Unit – 4 in all, of 18 marks each.

Course Contents :

UNIT-I

1. Origin and Meaning of Culture.
2. Culture, Civilization and the Individual

UNIT-II

3. Definition and evolution of the rational and irrational in Culture—The Role of Ideology and Utopia.

UNIT-III

4. Unity Vs. Plurality of Historical Cultures.

UNIT-IV

5. Concept of Tradition.
6. Sources of Indian Cultural Tradition.

Essential Readings :

1. Cassiurer, E. : *An Essay on Man* (N. York, Doubleday).
2. Sneh Pandit Ed. : *Perspectives on Philosophy of Culture* (Delhi, S. Chand, 1980).

OR

Paper IV Option (iii) : AESTHETICS

Max. Marks		: 100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hrs.

AIMS AND OBJECTIVES :
This paper aims at familiarizing the students with the meaning and scope of Aesthetics, art and society, aesthetic experience according to the Western and Indian tradition.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :
The syllabus has been divided into four units.
There shall be 9 questions in all. First question shall be short answer type containing 15 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions i.e. 2 marks of each. It shall carry 18 marks and shall be **compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall *attempt one question* from each Unit – 4 in all, of 18 marks each.

Course Contents :

UNIT-I

1. Aesthetics, its meaning and scope.
2. Nature and function of art.

UNIT-II

3. The relation of art to society.

UNIT-III

4. Meaning and purpose of art in the Indian Tradition.

UNIT-IV

5. Nature of Aesthetic experience according to the Western and Indian traditions.

Essential Readings :

1. Susan, L. Feagin and Patrick Maynard (ed.) : *Aesthetics,* Oxford University Press, 1997.
2. Hiriyanna : *Art Experience,* Karyalaya Publications, Mysore, 1954 .

PSYCHOLOGY

B.A. (HONOURS) THIRD YEAR EXAMINATION, 2011

OBJECTIVES :

I Its aim is to introduce the students history, importance and classification of psychological testing and provide them with knowledge and procedure of test construction and standardization.

II Pedagogy of the Course Work :

80% Lectures (including expert lectures).
20% assignments, discussion and seminars.

INSTRUCTIONS FOR PAPER-SETTER AND THE CANDIDATES :

There shall be 9 questions in all. Five questions in all are required to be attempted. The first question shall be Short Answer type containing 12 short questions spread over the whole syllabus and the candidate is required to answer 9 short questions in about 25-30 words each. It shall carry 18 marks and shall be **compulsory** question. Rest of the paper shall contain 4 Units. Each unit shall have two essay type questions and the candidates shall be given internal choice, i.e. the candidates shall attempt one question from each Unit – 4 in all, of 18 marks each.

Paper III : PSYCHOLOGICAL TESTING

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

UNIT-I : *Introduction to Psychological Tests :*

History of Psychological Testing; Nature of Psychological Tests.

Classification of Tests; Use of Tests.

UNIT-II : *Item Writing and Item Analysis :*

Characteristics of a Good Item; Various types of Items; Item Difficulty; Item Discrimination.

UNIT-III : *Reliability and Validity :*

Reliability : Concept, Types, Factors Affecting Reliability.

Validity : Concept, Types, Factors Affecting Validity; Relation between Reliability and Validity.

UNIT-IV : *Interpretation of Test Scores :*

Criterion Referenced Scores : Content Referenced and Outcome Referenced Scores; Norms Referenced Scores; Concept of Norms, Types of Norms (Percentile Scale and T. Scale).

Note : The use of non-programmable calculators and statistical tables is allowed in the examination.

Books Recommended :

Essential Readings :

1. Anastasi, A. (1998) : *Psychological Testing,* New York : Macmillan.
2. Gregory, R.J. (1996) *Psychological Testing,* Singapore : Allyn and Bacon.
3. Guilford, J.P. and Fruchter, B. (1981) : *Fundamental Statistics in Psychology and Education,* Singapore : McGraw Hill.
4. Guilford, J.P. (1954) : *Psychometric Methods,* New Delhi : Tata McGraw Hill.

Reference Books :

1. Freeman, F.S. (1962) : *Theory and Practice of Psychological Testing,* New Delhi : Oxford and IBH.
2. Cronbach, L.J. (1990) : *Essentials of Psychological Testing,* New York : Harper and Row.
3. Brown, F.G. (1976) : *Principles of Educational and Psychological Testing,* New York : Holt, Rinehart and Winston.

Paper IV : COUNSELLING PSYCHOLOGY AND ORGANIZATIONAL BEHAVIOUR

Max. Marks		: 100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

UNIT-I : Concept of Counselling; Approaches to Counselling; Freudian, Experiential (Client Centered, Gestalt, Existential), Behavioural Cognitive Behaviour Therapy.

UNIT-II : Process of Counselling, Introduction to Major Areas of Counselling : Educational, Vocational, Personal, and Family Counselling.

UNIT-III : Nature and History of Organization Behaviour.

Strategies of Organizational Development.

Communication : Nature, Models and Types.

UNIT-IV : *Work Motivation :* Early and Contemporary Theories.

Job Satisfaction : Nature and Factors.

Books Recommended :

Essential Readings :

1. Luthans, F. (1998) : *Organizational Behaviour,* New York : Irwin McGraw Hill.
2. Moorhead, G. and Griffin, R.W. (1999) : *Organizational Behaviour,* Delhi : AITBS.
3. Robbins, S.P. (1998) : *Organizational Behaviour,* New York : Prentice Hall of India.
4. Rao, S.N. (1992) : *Counselling and Guidance,* New Delhi : Tata McGraw Hill.
5. Nystul, M.S. (1999) : *Introduction to Counselling – An Art and Science Perspective,* Boston : Allyn and Bacon.

Reference Books :

1. Newstrom, J.W. and Davis K. (1998) : *Organizational Behaviour,* New Delhi : Tata McGraw Hill.
2. Woolfe, R., and Dryden, W. (Eds.) (1996) : *Handbook of Counselling Psychology,* New Delhi : Sage Publications.

GEOGRAPHY

B.A. (HONOURS) THIRD YEAR EXAMINATION, 2011

Paper- III : APPLIED GEOGRAPHY

Max. Marks	:	100
Terminal Exam.	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

Objectives :

To understand the prevalent issues in environment, society and economy and to provide a geographical interpretation with special reference to India.

Course Contents :

UNIT-I

Applied Geography : Meaning and concepts :

Meaning, nature, scope and significance of applied geography. Concept of planning region. Regionalisation scheme with reference to economic regionalisation.

UNIT-II

Industrial Location Theory of Weber and its application in industrial planning.

Central Place Theory and its application in location of central places.

Agricultural Location Theory and its application in agricultural location.

Growth Pole Theory and its application in regional development planning.

UNIT-III

Applied Geography : Method, techniques and contributions.

Cartographic techniques of dot, choropleth, graduated circle. Population and landuse mapping. Quantitative methods and techniques of crop combination; functional classification of urban centres.

UNIT-IV

Field surveys for rural and urban landuse.

Rural landuse studies; urban and regional planning; industrial planning; nature of resource management and planning.

Note :
1. A compulsory question containing15 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 10 parts in about 25-30 words each. Each part will carry 2 marks (total 20 marks).
2. The whole syllabus will be divided into 4 Units. Eight questions will be set out of the whole syllabus, 2 from each unit. The students will be required to attempt one question from each unit and each question will carry 17.5 marks. These will be in addition to the compulsory question at serial number 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

Books Recommended :

Essential Readings :

1. Carter, H. : *The Study of Urban Geography,* Arnold-Heinemann, New Delhi, 1979.
2. Chorley, R.J. & P. Haggett (Ed.) : *Models in Geography,* Methuen & Co., London, 1967.
3. Jones, Alun : *Fieldwork in Geography,* Longman, Green Co., London, 1968.
4. Stamp, L.D. : *Applied Geography,* Penguin Books, England, 1960.
5. Singh, R.L. (Ed.) : *Applied Geography, National Geographical Society of India,* Department of Geography, B.H.U., Varanasi, India.

Further Readings :

1. Dawson, J.A. : *Geography, Teach Yourself Books,* Hodder and Stoughton, U.K.
2. Misra, R.P. and Others : *Regional Development Planning in India,* Vikas, New Delhi.
3. Misra, V.C. & Others : *Essays in Applied Geography,* University of Sagar, India.
4. Sen Gupta, P. & Sadasyuk Galina : *Economic Regionalization of India, Problems and Approaches,* Registrar General Census of India.
5. Singh, Jasbir & Dhillon, S.S. : *Agricultural Geography,* Tata McGraw Hill, New Delhi, 1994.
6. World Report, World Bank & OUP (Annual Report)

Pedagogy :

- Exposure to media.
- National Five Year Plan documents and publications related to planning.
- Field work related to land–use and other socio-economic issues.

Paper – IV *: Any one of the following options :*

Option (i) : AGRICULTURAL GEOGRAPHY (An Elementary Course)
Option (ii) : POLITICAL GEOGRAPHY (An Elementary Course)
Option (iii) : ENVIRONMENTAL GEOGRAPHY (An Elementary Course)

Option (i) : AGRICULTURAL GEOGRAPHY (An Elementary Course)

Max. Marks	:	100
Terminal Exam.	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

Objectives :

- To familiarise the students with the concepts and development of agriculture.
- To examine the role of agricultural determinants towards changing cropping patterns, intensity, productivity, diversification and specialisation. The course further aims to familiarise the students with the application of various theories, models and classification schemes of cropping patterns and productivity.
- Its objective is also to discuss environmental, technological and social issues in agricultural sector with special reference to India.

Course Contents :

UNIT-I

Nature, scope and significance of agricultural geography.

General landuse classification.

Physical Determinants of Agricultural Landuse : Relief, climate, soils.

UNIT-II

Social and cultural determinants of agricultural landuse, land tenure, size and fragmentation of holdings and labour.

Economic Determinants of Agricultural Landuse : Marketing facilities, transport facilities, tariff and import restrictions, price incentives, credit.

UNIT-III

A critical review of Whittlesey's agricultural classification.

Characteristics and world patterns of the following :

Subsistence Types of Agriculture : Nomadic herding, shifting cultivation, rudimentary sedentary tillage, intensive subsistence tillage, subsistence crop and livestock farming.

UNIT-IV

Commercial Types of Agriculture : Livestock ranching, commercial plantation farming, commercial grain farming, mixed farming, commercial dairy farming.

Note :

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 10 parts in about 25-30 words each. Each part will carry 2 marks (total 20 marks).
2. The whole syllabus will be divided into 4 Units. Eight questions will be set out of the whole syllabus, 2 from each unit. The students will be required to attempt one question from each unit and each question will carry 17.5 marks. These will be in addition to the compulsory question at serial number 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidate(s) who have not been assessed earlier for Internal Assessment, the question paper(s) in their case shall be of Maximum Marks allotted to the paper(s) concerned.

Books Recommended :

Essential Readings :

1. Alexander, J.W. : *Economic Geography,* Prentice Hall, Englewood Cliffs, Latest Edition.
2. Husain, M. : *Systematic Agricultural Geography,* Inter-India Publications, New Delhi, 2000.
3. Singh, J. and Dhillon, S.S. : *Agricultural Geography,* Tata McGraw Hill Publishing, New Delhi, 1994.

Further Readings :

1. Morgan, W.B. and Munton, R.J.C. : *Agricultural Geography,* Methuen, London, 1971.

2. Symons, L. : *Agricultural Geography,* G. Bell & Sons, London, 1964.

Pedagogy :

The teacher should impress the students the overall importance of agriculture in the global perspective. The world is fast changing and its impact is felt on agriculture. Population is increasing and demand of agricultural products is also on the increase. Contrary to it, the farm lands are decreasing, that necessitates infusion of technology in agricultural sector. It is causing environmental pollution. The teacher should interact with students on said issues. Examples from neighbouring areas may be given to the students for better perceptions.

Option (ii) : POLITICAL GEOGRAPHY (An Elementary Course)

Max. Marks	:	100
Terminal Exam.	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

Objectives :

- To familiarise the students with the geographical factors which have a bearing on the political/administrative organisation of space.
- To enhance awareness among the students of multi-dimensional nature of geo-political space.

Course Contents :

UNIT-I

Nature and scope of political geography, phases in the development of political geography. Physical elements of political geography : Location, size and shape, relief, climate, seas and oceans.

UNIT-II

Economic Elements of Political Geography : Basic natural resources; food and power, minerals, industry, trade and transportation.

Human-Cultural Elements of Political Geography : Ethnographic structure of nations, language, religion, population, quantity and quality.

UNIT-III

Concepts of Frontiers and Boundaries : Classification of boundaries.

Global Distribution of Political Power : Views of Mackinder, Mahan and Spykman and their contemporary relevance.

UNIT-IV

International Tension : Identification of tension areas and factors contributing to tensions in different areas with special examples from Middle East and Indian Ocean.

Political Geography of the Indian Ocean.

Note :

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 10 parts in about 25-30 words each. Each part will carry 2 marks (total 20 marks).
2. The whole syllabus will be divided into 4 Units. Eight questions will be set out of the whole syllabus, 2 from each unit. The students will be required to attempt one question from each unit and each question will carry 17.5 marks. These will be in addition to the compulsory question at serial number 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidate(s) who have not been assessed earlier for Internal Assessment, the question paper(s) in their case shall be of Maximum Marks allotted to the paper(s) concerned.

Books Recommended :

Essential Readings :

1. De Blij, H.J., Glassner, M. : *Systematic Political Geography*, John Wiley, New York, 1968.
2. Dikshit, R.D. : *Political Geography : A Contemporary Perspective,* Tata McGraw Hill, New Delhi, 1996.

Further Readings :

1. Adhikari, Sudeepta : *Political Geography,* Rawat, Jaipur, 1997.
2. Muir, R. : *Modern Political Geography,* Macmillan, London, 1981.
3. Prescott, J.R.V. : *Political Geography,* Methuen, London, 1992.

4. Prescott, J.R.V. : *The Geography of Frontiers & Boundaries,* Aldine Pvt. Ltd., Chicago, 1965.

5. Valkenberg, S.V. and Stotz, Carl, L. : *Elements of Political Geography,* Prentice Hall of India, New Delhi, 1963.

Pedagogy :

- Field work to understand the political/administrative boundary configuration and people's problems and their perceptions.
- Consult political maps (large scale, small scale).
- Atlases and archival records.
- Collect relevant newspaper items for group discussion.
- Prepare pin-up board for display of important events of geopolitical nature.

Option (iii) : ENVIRONMENTAL GEOGRAPHY (An Elementary Course)

Max. Marks	:	100
Terminal Exam.	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

Objectives :

- To understand the significance of environment and its interrelationship with man.
- To understand the importance of soil and factors responsible for soil erosion.
- The importance of energy sources, forest and wildlife and their preservation to be studied.

Course Content :

UNIT-I

Environment and Geography : Relationship between geography and environment; definition and scope of environmental geography; significance of environmental studies.

The Environment : Characteristics, components and functioning; concept of healthy environment.

(6+6 = 12 Lectures).

UNIT-II

The Soil System : Origin and composition; horizons, texture, chemistry, factors influencing soil types; soil erosion, soil conservation; and soil pollution.

Energy : Energy flow in the ecosystem; environmental impact of both conventional and non-conventional energy sources. (6+4 = 10 Lectures)

UNIT-III

Forests : Significance; world reserves; deforestation; and conservation strategies.

Wildlife : Wildlife and environment relationship, regional distribution; conservation of wildlife; and wildlife in India.

(5+5 = 10 Lectures)

UNIT-IV

Population and Environment : Basic issues, environmental impact of population growth and agenda for action. Global Environmental Issues : Water contamination; atmospheric pollution; acid rain; noise pollution; and control strategies.

Legislative strategies, environmental issues and the international community; Rio-de-Janeiro Earth Summit of 1992; selected environmental protection laws in India; Environmental Protection Act, National Environmental Tribunal Act, Wildlife Protection Act, Forest Conservation Act, Air Act, Water Act.

(6+6 +6 = 18 Lectures)

Note :

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 10 parts in about 25-30 words each. Each part will carry 2 marks (total 20 marks).

2. The whole syllabus will be divided into 4 Units. Eight questions will be set out of the whole syllabus, 2 from each unit. The students will be required to attempt one question from each unit and each question will carry 17.5 marks. These will be in addition to the compulsory question at serial number 1.

3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

4. For reappear/improvement candidate(s) who have not been assessed earlier for Internal Assessment, the question paper(s) in their case shall be of Maximum Marks allotted to the paper(s) concerned.

Books Recommended :

Essential Readings :

1. Adam, W.M. : *Green Development : Environment & Sustainability in the Third World*, Routledge, London, 1990.

2. Chandna, R.C. : *Environment Awareness,* Kalyani Publishers, New Delhi, Clarendon Press, Oxford, 1998. Also available in Hindi and Punjabi.

3 Chandna, R.C. : *Environmental Geography,* Kalyani Publishers, New Delhi, 2001.

4. Goudie, A. S. : *The Human Impact on Environment,* Blackwell, Oxford, 1993.

Further Readings :

1. Ehrlich Paul, R. et al : *Ecoscience – Population, Resources and Environment,* Freeman and Co., San Francisco, 1977.

2. Goudie, A. S. : *Environmental Change, Contemporary Problems in Geography,* Clarendon Press, Oxford, 1992.

3. Pickering, K.T. : *An Introduction to Global Environmental Issues,* Routledge, London, 1994.

4. Singh, Savinder : *Environmental Geography,* Paryag Pustak Bhawan, Allahabad, 1991.

5. Strahler, A. N. & Strahler, L. A. : *Geography and Man's Environment,* John Wiley & Sons, New York.

Pedagogy :

- The interaction of teacher and students is must on issues related to environment, energy resources, forest and soil preservation.
- A field visit to those areas where soil erosion is a serious problem be arranged and methods of conservation be taught.

PUBLIC ADMINISTRATION

B.A. (HONOURS) THIRD YEAR EXAMINATION, 2011

Outline of Tests, Syllabi and Courses of Reading

Paper III : RESEARCH METHODS AND STATISTICS

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

INSTRUCTIONS FOR PAPER-SETTERS AND CANDIDATES :

- For Private/University School of Open Learning (USOL) students, who have not been assessed earlier for the internal assessment, the marks secured by them in the paper will be proportionately increased in lieu of the internal assessment.

  The Paper-Setter must put a note in question paper in this regard.

- The candidate shall attempt 5 questions in all (one compulsory and one each from four units). The first **compulsory** question shall comprise of 12 short-answer type questions, covering the whole syllabus, to be answered in 25-30 words each, out of which the candidate would be required to attempt any 9. Each question will carry 2 marks. Rest of the paper shall contain 4 units, each unit having two questions, out of which the candidate would be required to attempt one question from each unit. Each question will carry 18 marks.

UNIT-I

Meaning, Nature and Scope of Social Research.

Concepts; Hypotheses : Meaning, Types and Significance.

Research Design—Meaning, Types and Components.

Method of Social Research—Experimental–Survey and Case Study.

UNIT-II

Methods of Data Collection—Documents, Observation, Interview and Questionnaire.

Data Processing—Editing, Coding and Tabulation, The use of Computer.

Report Writing.

UNIT-III

Sampling—Meaning, Types, Merits and Demerits.

Frequency Distribution – Meaning and Steps.

Measures of Central Tendency – Mean, Median and Mode.

UNIT-IV

Measures of Dispersion –Standard Deviation.

Correlation – Karl Pearson.

Chi-Square

Essential Readings :

1. C.A. Moser and Kalton, G. : *Survey Methods in Social Investigations* (London : ELBS, 1979).
2. C. Seltiz & Others : *Research Methods in Social Relations* (New York : Rinehart & Winston, 1964).
3. S.P. Gupta : *Statistical Methods* (New Delhi : Sultan Chand, Rev. ed. 1987).
4. W.J. Goode and P.K. Hatt : *Methods in Social Research* (U.S.A. : McGraw Hill, 1990).

Further Readings :

1. Arthur Bowley *Elements of Statistics* (New York : Staples, 6th Ed., 1947).
2. Ghosh and Chowdhry : S*tatistics* (Allahabad : Indian Press, Rev. Enlarg. Ed., 2004).
3. L. Cohen : *Statistical Method for Social Scientists* (New Delhi : Prentice Hall of India Private Ltd.) (1968).

PAPER IV : *The candidates have to select one out of the following four options :*

Option (i) : Organisation Behaviour.

Option (ii) : Cooperative Administration in India.

Option (iii) : Public Enterprise Management (with special reference to India).

Option (iv) : Environment Administration in India.

Paper IV : Option (i) : ORGANISATION BEHAVIOUR

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

INSTRUCTIONS FOR PAPER-SETTERS AND CANDIDATES :

- For Private/University School of Open Learning (USOL) students, who have not been assessed earlier for the internal assessment, the marks secured by them in the paper will be proportionately increased in lieu of the internal assessment.

  The Paper-Setter must put a note in question paper in this regard.

- The candidate shall attempt 5 questions in all (one compulsory and one each from four units). The first **compulsory** question shall comprise of 12 short-answer type questions, covering the whole syllabus, to be answered in 25-30 words each, out of which the candidate would be required to attempt any 9. Each question will carry 2 marks. Rest of the paper shall contain 4 units, each unit having two questions, out of which the candidate would be required to attempt one question from each unit. Each question will carry 18 marks.

UNIT-I

Nature of Organisation Theory : Organisation Theory and other theories. Various approaches to Organisation Theory :

(i) Classical Organisation Theory – Basic Features and Shortcomings.

(ii) Neoclassical Organisation Theory – Elements and Limitations.

(iii) Modern Organisation Theory.

Nature of Human Behaviour.

UNIT-II

Psychological Foundations of Human Behaviour.

Motivation –Theories.

Attitudes.

Morale—Meaning, Causes and Remedies.

UNIT-III

Socio-culture Factors and Behaviour.

Leadership : Theories, Style and Qualities.

UNIT-IV

Communication.

Organisation Change and Development.

Decision making.

Essential Readings :

1. Avasthi and Maheshwari : *Public Administration,* Agra : Laxmi Narain Aggarwal, 2008.
2. L.M. Prasad : *Organisational Behaviour,* New Delhi : Sultan Chand, 3rd Rev. and Enlarg (Ed.), 2000.
3. K.K. Sharma & Pardeep Sahni : *Organisation Behaviour,* New Delhi : Deep & Deep.

Further Readings :

1. Robert Dubin : *Human Relations in Administration,* New Delhi : Prentice Hall, 3rd ed.
2. M. Smith : *Industrial Psychology,* New York : Philosophical Library.
3. B.J. Kolasa : *Introduction to Behavioural Sciences for Business.*
4. N.R.F. Maier : *Psychology in Industry* (New Delhi : Oxford, 3rd Edition).
5. B.Von Haller Gilmer : *Industrial and Organizational Psychology.*

Paper IV Option (ii) : CO-OPERATIVE ADMINISTRATION IN INDIA

Max. Marks		: 100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

INSTRUCTIONS FOR PAPER-SETTERS AND CANDIDATES :

- For Private/University School of Open Learning (USOL) students, who have not been assessed earlier for the internal assessment, the marks secured by them in the paper will be proportionately increased in lieu of the internal assessment.

  The Paper-Setter must put a note in question paper in this regard.

- The candidate shall attempt 5 questions in all (one compulsory and one each from four units). The first **compulsory** question shall comprise of 12 short-answer type questions, covering the whole syllabus, to be answered in 25-30 words each, out of which the candidate would be required to attempt any 9. Each question will carry 2 marks. Rest of the paper shall contain 4 units, each unit having two questions, out of which the candidate would be required to attempt one question from each unit. Each question will carry 18 marks.

UNIT-I

Co-operation : Meaning, Significance and Evaluation.

Co-operative Principles.

Co-operative Legislation : G.O.I. Act 1912 and Punjab Co-operative Societies Act, 1961 (as amended from time to time).

UNIT-II

Registration of a Co-operative Society.

General Body, Board of Directors and Committees.

Chief-Executive : Appointment, Functions and Role.

Co-operative Administration at Union Level – Organisation and Functions.

UNIT-III

Co-operative Department at State Level – Organisation and functions.

Registrar Co-operative Societies : Appointment and Role.

Forms of Co-operatives : Role of Apex Federations and Primary Co-operatives.
Co-operative Education.

UNIT-IV

Human Resource Management : Recruitment and Training.
Financial Management.

Essential Readings :

1. P.R. Dubhashi : Principles and Philosophy of Cooperation, Pune, VMNICM, 1970.
2. B.B. Goel : Dynamics of Cooperative Administration, Delhi, Deep and Deep, 1988.

3.	Goel, S.L. &. Goel, B.B.	:	*Principles, Problems and Prospects of Coop. Admn.,* Delhi, Sterling, 1979.
4.	Kamra, P.K.	:	*Cooperative Management,* Delhi : Deep and Deep.
5.	Krishnaswami, O.R.	:	*Fundamentals of Cooperative,* Delhi, S. Chand and Co., 1978.
6.	Mathur, B.S.	:	*Cooperation in India,* Agra, Sahitya Bhawan, 1971.
7.	Singh, Avtar	:	*Cooperative Law.*

Further Readings :

1.	Dwivedi, R.C.	:	*Democracy in Cooperative Movement.*
2.	Gill, M.S	:	*Agricultural Cooperatives,* Delhi, Vikas, 1983.
3.	I.C.A.	:	*Experts Consultation on Cooperative Member Education,* Delhi, ICA, 1980.
4.	I.C.A.	:	*Readings in Cooperative Management,* Delhi, ICA, 1977.
5.	I.L.O.	:	*Cooperative Management and Administration,* Geneva, ILO.
6.	Puri, S.S.	:	*Ends and Means of Cooperative Development,* Delhi, NCUI, 1979.
7.	Bhatia, B.S. (ed.)	:	*Encyclopaedia of Cooperative Management,* Lyall Book Depot.

Paper-IV Option (iii) : PUBLIC ENTERPRISE MANAGEMENT (with special reference to India)

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

INSTRUCTIONS FOR PAPER-SETTERS AND CANDIDATES :

- For Private/University School of Open Learning (USOL) students, who have not been assessed earlier for the internal assessment, the marks secured by them in the paper will be proportionately increased in lieu of the internal assessment.

  The Paper-Setter must put a note in question paper in this regard.

- The candidate shall attempt 5 questions in all (one compulsory and one each from four units). The first **compulsory** question shall comprise of 12 short-answer type questions, covering the whole syllabus, to be answered in 25-30 words each, out of which the candidate would be required to attempt any 9. Each question will carry 2 marks. Rest of the paper shall contain 4 units, each unit having two questions, out of which the candidate would be required to attempt one question from each unit. Each question will carry 18 marks.

UNIT-I

Public Enterprise : Meaning and Evolution.

Public Enterprises in India : Rationale and Role.

Administrative and Managerial Problems of Public Enterprises.

UNIT-II

Organisational Patterns : Departmental Undertakings, Public Corporation and Government Company.

Board of Directors : Rationale, Constitution, Types and Functions.

UNIT-III

Legislative Control over Public Enterprises.

Government Control over Public Enterprises.

Performance of Public Enterprises.

UNIT-IV

Public Enterprise Reforms since 1991.

Memorandum of Understanding.

Privatization : Meaning, Rationale, Achievements and Failures.

Essential Readings :

1. B.P. Mathur : *Public Enterprise Management* (Macmillan, New Delhi, 1999).
2. Jagdish Parkash : *Administration of Public Enterprises in India* (Himalaya Publishing House, Mumbai, 2002).
3. Laxmi Narain : *Public Enterprise Management and Privatisation* (S. Chand, New Delhi, 2005).
4. N.K. Sengupta : *Government and Business* (Vikas, New Delhi).

5. Ruddra Dutt and K.P.M. Sundaram : *Indian Economy* (S. Chand, New Delhi, 2009).

6. S.S. Khera : *Government in Business* (Delhi : National, 1977).

Further Readings :

1. A.H. Hanson : *Public Enterprises and Economic Development* (London : Routledge and Kegan, 1972).

2. K.R. Gupta : *Issues in Public Enterprises* (New Delhi, S. Chand, 1975).

3. Government of India : *Public Enterprises Survey,* Ministry of Heavy Industry & Public Enterprises, Vol. 1, 2, 3 DPE, New Delhi.

4. Ratnakar Gedam : *Public Sector Management* .

5. United Nations : *Countries* (New York : V.N., 1974).

6. V.V. Ramanadham : *The Structure of Public Enterprises and Economics in India* (Bombay, Asia Publishing House).

7. V.V. Ramanadham : *The Working of Public Sector* (Bombay : Allied Publishers).

Paper IV Option (iv) : ENVIRONMENT ADMINISTRATION IN INDIA

Max. Marks		: 100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

INSTRUCTIONS FOR PAPER-SETTERS AND CANDIDATES :

- For Private/University School of Open Learning (USOL) students, who have not been assessed earlier for the internal assessment, the marks secured by them in the paper will be proportionately increased in lieu of the internal assessment.

  The Paper-Setter must put a note in question paper in this regard.

- The candidate shall attempt 5 questions in all (one compulsory and one each from four units). The first **compulsory** question shall comprise of 12 short-answer type questions, covering the whole syllabus, to be answered in 25-30 words each, out of which the candidate would be required to attempt any 9. Each question will carry 2 marks. Rest of the paper shall contain 4 units, each unit having two questions, out of which the candidate would be required to attempt one question from each unit. Each question will carry 18 marks.

UNIT-I

Environment – Concept and Components.

Environment Administration – Meaning, Nature and Scope.

Sustainable Development – Concept, Issues and Strategy.

UNIT-II

Environment Problems and Threats–Air, Water, Land, Human Settlements and Forests.

Impact of Development, Urbanization and Industry on Environment.

Organisation and Machinery of the Government at the Union Level for Environment Protection.

Organisation and Machinery of the Government at the State Level for Environment Protection.

UNIT-III

Pollution Control Boards–Organisation, Functions and Role.

Environmental Policy with special reference to Water, Air and Forest; Constitutional Provisions; Legislative Measures.

Environmental Priorities in India—Population Stabilisation; Forest Cover; Deforestation; Wildlife Protection; Recycling of Wastes.

UNIT-IV

Environment and Global Issues—Conventions; Depletion of Ozone Layer; Global Warming.

Environment Education, Awareness and Ethics; Role of NGOs in Environment Protection and Conservation.

Essential Readings :

1. R.C. Chandana : *Environment Awareness* (New Delhi : Kalyani, 1998).
2. R.K. Sapru : *Development Administration* (New Delhi : Sterling, 2002).
3. R.K. Sapru : *Public Policy* (New Delhi : Sterling, 2004).
4. S. Edmunds and J. Latey : *Environmental Administration* (New York : McGraw-Hill, 1973).
5. V.K. Nanda : *Environment Education* (New Delhi : Anmol).

Further Readings :

1. India, Centre for Science and Development : *The State of India's Environment : A Citizen's Report* (New Delhi : CBSE).

2. Kumar, Naresh : *Air Pollution and Environment Protection : Legislative Policies, Judicial Trends and Social Perceptions* (New Delhi : Mittal, 1999).

3. Sheo Parkash Agnihotri : *Environment Conservation Management and Planning* (Allahabad : Chugh, 1992.)

4. Radha, S. and Sankhyan, A.S. : *Environmental Challenges of the 21st Century* (New Delhi, Deep, 2004).

5. Monga, G.S. : *Environment and Development* (New Delhi : Deep and Deep, 2004).

ANCIENT INDIAN HISTORY, CULTURE & ARCHAEOLOGY

B.A. (HONOURS) THIRD YEAR EXAMINATION, 2011

Note : The students shall have to opt for two papers for third year examination out of the following four papers :

Outlines of Test :

Paper I	:	Archaeology
Paper II	:	Epigraphy and Numismatics
Paper III	:	History of Art and Architecture (Earliest times to C. 1000 A.D.)
Paper IV	:	Spread of Indian Culture in South-East Asia upto 12th Century A.D.

Paper-I : ARCHAEOLOGY

Total Marks	:	100
Theory	:	90
Internal Assessment	:	10
Time	:	3 Hours

Objectives :

This course is a survey of the pre and proto-historic background to Indian history from the beginning of man to the early historical period. Excavated sites are studied for an in site into the growth of early Indian culture.

Note :

1. The paper setter is required to set 9 questions in all.
2. The candidates are required to attempt five questions in all.
3. All questions shall carry equal marks.
4. The first question shall be short answer type containing 15 short questions spread over the entire syllabus. The candidates are required to attempt 9 short questions in about 25 to 30 words each. It shall carry 18 marks and shall be a **compulsory** question.
5. The rest of the paper shall contain 4 Units. The entire syllabus has been divided into 4 Units. Each unit shall have two questions and the candidate shall be given internal choice i.e. the candidate shall attempt one question from each Unit i.e. 4 in all, of 18 marks each.

UNIT-I : The Palaeolithic, Mesolithic and Neolithic cultural sequences in various parts of the Indian subcontinent.

UNIT-II : (a) The background and the phenomenon of the Harappan civilization.

(b) The Chalcolithic cultures of western and southern India.

UNIT-III : The beginning of iron technology and the Painted Grey Ware Culture.

UNIT-IV : Excavations of early historic cities of :

(a) Northern India and Gangetic doab : Taxila and Hastinapura.

(b) Eastern India : Sisupalgarh and Chandraketugarh.

(c) Southern India : Arikamedu and Brahmagiri and Chandravalli.

Pedagogy of the Course Work :

The students are made familiar with prehistoric, proto historic and historical sites through visits. Slides, photographs of relevant sites, their significance in establishing chronologies is also discussed in the class. Visits to museums is also an important aspect through which students are made familiar with the material evidence.

Paper-II : EPIGRAPHY AND NUMISMATICS

Total Marks	:	100
Theory	:	90
Internal Assessment	:	10
Time	:	3 Hours

Objectives :

The paper is aimed to prepare students to handle the original source material – inscriptions and coins, which make the backbone of Ancient Indian History, as without the knowledge of epigraphy and numismatics, the subject cannot be mastered.

Note :

1. The paper setter is required to set 9 questions in all.
2. The candidates are required to attempt five questions in all.
3. All questions shall carry equal marks.
4. The first question shall be short answer type containing 15 short questions spread over the entire syllabus. The candidates are required to attempt 9 short questions in about 25 to 30 words each. It shall carry 18 marks and shall be a **compulsory** question.
5. The rest of the paper shall contain 4 Units. The entire syllabus has been divided into 4 Units. Each unit shall have two questions and the candidate shall be given internal choice i.e. the candidate shall attempt one question from each Unit i.e. 4 in all, of 18 marks each.

UNIT-I

(a) Origin and antiquity of writing in India with special reference to the origin of Brahmi script.

(b) Origin and antiquity of coinage in India.

UNIT-II

(a) Asokan inscriptions – Pillar Edicts I to IV; Girnar Rock Edicts I, II and XIII. (Both for study and decipherment).

(b) Besnagar Pillar inscription of Heliodorus; Nasik Cave Inscription of Queen Balasri; Sarnath Buddha Image inscription of Kanishka's year 3; Junagarh Rock inscription of Rudradaman I, Saka year 72. (Study only). Gupta inscriptions—Allahabad Pillar inscription of Samudragupta, Mehrauli Iron Pillar inscription of Chandra, Bhitari Stone Pillar inscription of Skandagupta (Study only).

(c) Post-Gupta inscriptions : Mandasor Stone inscription of Malava Samvat 589, Banskhera Copper-plate of Harshavardhana, Pehowa inscription of Bhojadeva (Study only).

UNIT-III

(a) Coins as source of ancient Indian history.

(b) Study of Punch marked coins.

UNIT-IV

(a) Coins of the Kushanas.

(b) Coins of the Guptas.

Pedagogy of the Course Work :

Students should be familiarized with the details of the topics in the syllabus with stress on the contents of each topic. Visual aids like the photographs/slides of the coins and inscriptions should be used regularly. The decipherment of Asokan inscriptions should be taught with the help of plates and paleographic charts.

Paper III : HISTORY OF ART AND ARCHITECTURE (EARLIEST TIMES TO C. 1000 A.D.)

Total Marks	:	100
Theory	:	90
Internal Assessment	:	10
Total	:	3 Hours

Objectives :

The main objective of the paper is to acquaint the students with the architecture of ancient Buddhist and Hindu monuments, stone and bronze sculptures, that form an integral part of ancient history and heritage, for comprehensive understanding of the subject.

Note : 1. The paper setter is required to set 9 questions in all.

2. The candidates are required to attempt five questions in all.

3. All questions shall carry equal marks.

4. The first question shall be short answer type containing 15 short questions spread over the entire syllabus. The candidates are required to attempt 9 short questions in about 25 to 30 words each. It shall carry 18 marks and shall be a **compulsory** question.

5. The rest of the paper shall contain 4 Units. The entire syllabus has been divided into 4 Units. Each unit shall have two questions and the candidate shall be given internal choice i.e. the candidate shall attempt one question from each Unit i.e. 4 in all, of 18 marks each.

UNIT-I

(a) Harappan architecture – Town planning and important buildings.

(b) Harappan sculptures – Stone and bronze.

UNIT-II

(a) Stupa Architecture – Salient features, Stupas of Sanchi and Amaravati.

(b) Rock-cut Architecture–Salient features, Lomas Rishi Cave, Ajanta and Ellora.

UNIT-III : Temple Architecture :

(a) Gupta temples.

(b) Nagara – Khajuraho and Orissa.

(c) Dravida – Pallava and Chola.

(d) Vessara – Chalukya.

UNIT-IV : Sculpture :

(a) Early Period – Mauryan, Sunga and Kushanas.

(b) Gupta and Early Medieval – Gupta, Pala and Chola.

Pedagogy of the Course Work :

Besides giving an outline of the history and salient features of the topics enumerated in the syllabus, students should be thoroughly acquainted with important monuments and images in stone and bronze of each period. Important schools of art should also be taught. Constant help of visual aids like slides and transparencies should form integral part of teaching. Field work trips to museums and sites of monuments should be undertaken.

Paper-IV : SPREAD OF INDIAN CULTURE IN SOUTH-EAST ASIA UPTO 12TH CENTURY A.D.

Total Marks	:	100
Theory	:	90
Internal Assessment	:	10
Time	:	3 Hours

Objectives :

This course enlightens the students about the spread of Indian culture in Burma, Thailand, Indonesia, Combodia and Champa. In order to teach the students about the elements of Indian Culture, which made known its impact in those countries, it is, therefore, necessary to discuss the political history, art and architecture and religion of those regions.

Note :

1. The paper setter is required to set 9 questions in all.
2. The candidates are required to attempt five questions in all.
3. All questions shall carry equal marks.
4. The first question shall be short answer type containing 15 short questions spread over the entire syllabus. The candidates are required to attempt 9 short questions in about 25 to 30 words each. It shall carry 18 marks and shall be a **compulsory** question.
5. The rest of the paper shall contain 4 Units. The entire syllabus has been divided into 4 Units. Each unit shall have two questions and the candidate shall be given internal choice i.e. the candidate shall attempt one question from each Unit i.e. 4 in all, of 18 marks each.

UNIT-I

(a) Beginning of Indianisation in Burma.

(b) Art and Religion of Burma.

UNIT-II

(a) Beginning of Indianisation and early history of Thailand or Siam.

(b) Beginning of Indianisation in Java and establishment of Hindu Colonies.

(c) Study of Indo-Japanese Art.

UNIT-III

(a) Beginning of Indianisation in Combodia and Champa.

(b) Art of Combodia and Champa.

UNIT-IV

(a) Beginning of Indianisation in Indonesia.

(b) Study of Art and Archaeological remains of Indonesia.

Pedagogy of the Course Work :

The students are taught with the help of slides, photographs and maps. In addition to it, lectures, workshops, seminars, written assignments and class discussions, term papers etc. are essential parts of it. The students should be made familiar with the outline of the topics mentioned in above units.

MUSIC (VOCAL AND INSTRUMENTAL)

B.A. (HONOURS) THIRD YEAR EXAMINATION, 2011

Paper-III : (Theory)

Total Marks	:	100 Marks
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Total	:	3 Hours

Note : There will be 10 questions in all and a student will attempt five questions choosing at least one from each unit. Unit V will be compulsory and that will be of 22 marks. Unit V of the question paper will be based on Unit V of the syllabus. Remaining four questions will be of 17 marks each.

UNIT-I

1. Definition with full explanation of the following :

   Sthai, Antra, Krintan, Badhat, Jod-Alap, Lag Daant, Jugalbandi, Layakaries (Adi, Kuadi, Biadi), Ghaseet, Meend, Kan, Khatka, Murki, Kalavant, Vaggeykar.

UNIT-II

1. Concept of Aesthetics in Music.
2. Definition and Principles of Aesthetics of Music and its Importance.

UNIT-III

1. Topics of Essay :
   (i) Music and Fine Art.
   (ii) Modern Methods of Teaching Music.
   (iii) Sangeet Sammelan.
   (iv) Role of Music in National Integration.
   (v) Lok Sangeet.

UNIT-IV

1. Knowledge of musical aptitude and musical talent.
2. Knowledge of good qualities of a music performer (Vocalist or an instrumentalist)

UNIT-V

1. Description of Detailed & Non-detailed Ragas :

   Detailed Raga :

   (i) Marwa.

   (ii) Madhmad Sarang.

   (iii) Patdeep.

   (iv) Mian Ki Malhar.

   (v) Poorvi

   Non-detailed Ragas :

   (i) Sohni.

   (ii) Megh.

   (iii) Tilang

2. Notation of Khayals, Dhrupad and Dhamar in ragas prescribed for B.A. (Honours) III Years course.

Paper-IV : Stage Performance

Total Marks	:	100 Marks
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	45 minutes

Part-A : Stage Performance (Choice) 45 Marks

(a) A candidate has to demonstrate in the Raga of his/her choice with detailed elaboration :

(i)	Vilambit Khayal	:	15 marks	= 25 marks
(ii)	A Drut Khayal	:	10 marks	

OR

(i)	A Vilambit Gat	:	15 marks	= 25 marks
(ii)	A Drut Gat	:	10 marks	

(b)	One Dhamar in Gayaki Ang	:	10 marks	= 10 marks
	OR			
	One Drut Gat in Ek-Tal	:	10 marks	
(c)	Gazal, Geet, Bhajan, Shabad	:	5 marks	= 5 marks
	OR			
	One Dhun on Instrument	:	5 marks	
(d)	Tuning of the Instrument	:		= 5 marks

Part-B : Practical (Viva) 45 Marks

(FOR VOCAL MUSIC)

(a) Two Vilambit Khayals in any two of the prescribed Ragas with proper Gayaki.

(b) One Drut Khayal with proper Gayaki in each of the prescribed ragas.

(FOR INSTRUMENTAL MUSIC)

(a) Two Vilambit Gats in any two of the prescribed Ragas with elaboration and style.

(b) One Drut Gat with Alap, Jod, Toras and Jhala in each of the prescribed ragas.

(c) Critical study of the Ragas prescribed for Practical exam.

List of Detailed Ragas and Talas :

Ragas : Marwa, Madhumad Sarang, Patdeep, Mian-Ki-Malhar and Poorvi
Non-Detailed Ragas : Megh, Tilang, Sohni.

Talas : Tivra, Sultal, Tilwada, Ada-Chautal and Deepchandi, Demonstration both by hands and by playing on Tabla.

Note : The examiner can ask comparative study of detailed and non-detailed ragas.

HOME SCIENCE

The syllabus of Home Science for B.A./B.Sc. (Honours) Third year has been **KEPT IN ABEYANCE.**

PHYSICS

Note : There will be following two papers :

Paper-III : Digital Electronics and Material Science	100	:	Marks
	Time	:	3 Hrs.
Paper-IV : Practicals	100	:	Marks

Note : The paper will consist of three sections :

Section A : 4 Questions

Section B : 4 Questions

Section C : 1 Question

Contents of Section A and B are indicated in the syllabus for the paper. Section C will consist of one question carrying 10 parts of short answer type covering the whole syllabus. The candidate will attempt the compulsory question and two questions each from Section A and B, five questions in all. All questions carry equal marks.

PAPER-III : Digital Electronics and Material Science

Section A : Digital Electronics

1. *Number Systems :*
   Binary, Hexadecimal and octal number systems, Interconversions and arithmatic operations, Binary fractions and negative binary numbers, floating point representations. Problems 2. 1 to 2.37 of Reference 1.

2. *Binary Codes :*
   BCD 8421. XS3, Gray codes, Error detecting and correcting codes. : Problems 3.1 to 3.18 of Reference 1.

3. *Boolean Algebra :*
   AND, OR, NOT, Exclusive OR operations, Boolean identities, De-Morgan's theorem and simplification of Boolean functions, NAND, NOR Gates, Negative Logic Problems 4.8 to 4.11, 4.17 to 4.26 of Reference 1.

4. *Logic Hardware :*
   Diode, BJT, FET, Switching devices, Logic specifications, Logic families and their performance.

5. *Combinational Logic :*
   Minterms and Maxterms, Mapping POS, SOP forms, Examples Decoders, Code convertors, full adders. Problems 6.1 to 6.20 of Reference 1.

Reference Books

1. *Digital Electronics : An Introduction to Theory and Practice* by W.H. Gothmann, Prentice Hall of India Pvt. Ltd.

2. *Introduction to Digital Electronics* by Dr. Mohinder Singh, Kalyani Publishers, New Delhi.

Section B : Material Science

1. *Magnetic Materials :*
Classification, Dia and para magnetism, Ferro and ferrimagnetism, Two-sublattice models. Ferrimagnetic materials. Exchange interaction in magnetic materials, Magnetic anisotropy and magnetic-striction, magnetic domain–origin, size and hysteresis, Hard and Soft magnetic materials, Applications of permanent magnets. Ferrites for microwave application. Magnetic bubbles. (13.1, 13.2, (3.2.1) 13.3 – 13.9 of Book).

II. *Dielectric Materials :*
Dielectric constant and polarizability – Electronic, Ionic and dipolar polarizabilities, Frequency dependence of polarizability, Dielectric loss, Insulating materials, Ferroelectrics, Piezoelectrics, Electrets, Measurement of dielectric properties, Applications (14.1 – 14.8 of Book).

III. *Polymers :*
Classification Structure – Property correlation, Molecular weight, Crystallinity in polymers, Mechanical properties, Applications (16.1 – 16.6 of Book).

IV. *Materials for Nuclear Engineering :*
Nuclear fission, Nuclear reaction cross-section, Types of nuclear reactors, Nuclear fuel, uranium enrichment fuel elements. Fuel cladding, Moderators, Control materials, Coolants, Shielding materials (17.2 – 17.11 of Book).

V. *Some Techniques for Material Studies :*
Optical microscopy, Electron microscope Chemical Analysis using atomic absorption spectroscopy, Photoelectron spectroscopy, X-ray fluorescence and Auger processes (20.1– 20.4 of Book).

Book :

Science of Engineering Materials, C.M. Srivastava and C. Srinivasan (Wiley Eastern, New Delhi, 1991).

PAPER IV : Physics Practicals

Any Ten of the following experiments :

1. Study of amplitude response of a forced oscillator to the damping.

2. Study of coupling factor between two pendulums in its dependence on coupling mass and distance of coupling threads from the axis of oscillation.

3. Viscosity of air by rotating cylinder method.

4. Determination of Stefan's constant.

5. Bi-prism-determination of wavelength and thickness of a thin transparent sheet.

6. Verification of Fresnel's laws of reflection.

7. Fresnel's diffraction at a single slit using laser beam.

8. Characteristics of given voltage doubler and tripler.

9. Conductivity of a semi conductor using four-probe method.

10. Hall effect.

11. G.M. Counter – Statistical fluctuations.

12. Characteristics of lc. 555.

13. Stray capacitance of a coil.

14. To verify that the power dissipation in the side bands is directly proportional to the square of the modulated power.

15. To study various biasing circuits for transistor.

16. To verify Boolean identities.

CHEMISTRY

B.Sc. (HONOURS) THIRD YEAR EXAMINATION, 2011

Outlines of Tests

				Marks	
			Lecture Hours	*Theory*	*Internal Assessment*
Paper III	:	Inorganic Chemistry	40	90	10
Paper IV	:	Physical Chemistry	40	90	10

Paper III : INORGANIC CHEMISTRY

Total Marks	:	100
Theory	:	90
Internal Assessment	:	10
Total Lectures	:	40

OBJECTIVE OF THE COURSE :

To teach the fundamental concept of Chemistry and their applications, the syllabus pertaining to B.Sc. (Honours) in the subject of Chemistry has been upgraded as per provision of the UGC module and demand of the academic environment. The course contents have been revised from time to time as per suggestions of the teachers of the Chemistry working in the Panjab University, Chandigarh and affiliated colleges. The syllabus contents are duly arranged unit wise and contents are included in such a manner so that due importance is given to requisite intellectual and laboratory skills.

UNIT-I

1. **Inorganic Chains, Rings and Cages :** (10 lectures)

   (a) *Chains :* Catenation including heterocatenation, silicate minerals, intercalation, One dimensional sulphur-nitrogen conductor.

   (b) *Rings :* Borazines, phosphazenes and phosphazene polymers.

   (c) *Cages :* Simple cage molecules; Phosphorus–oxygen and phosphorus-sulphur cages; introduction to carboranes and metallocarboranes.

UNIT-II

2. **Organometallics :** (10 lectures)

   Formation and reactions (i) carbonylate ions (ii) metal-carbonyl hydrides & (iii) metal carbonyl halides.

   Transition metal-carbon multiple bonds : Chemistry of :

   Alkylidene Complexes.

   Alkylidyne complexes including bridged molecules.

   Metallocumulenes.

Unit-III

(10 lectures)

Complexes of aromatic systems with transition metals. M.O. description of bonding in Ferrocene.

Study of coordinative unsaturation, oxidative addition, and insertion reactions in Organometallics.

Phenomenon of Fluxionality in Organometallics and its characterization.

Unit-IV

3. **Reaction Mechanism :** (10 lectures)

Nature and classification of lignad substitution reactions, application of V.B. & C.F. theories to predict the substitution reactions. Mechanism of ligand replacement reactions, substitution in octahedral complexes acid hydrolysis, base hydrolysis, acid-catalysed, acid hydrolysis and acid hydrolysis for chelates. Substitution in square planar complexes. The trans-effect & its Synthetic applications, M.O. theories and trans-effects. Electron transfer reactions & their mechanism. Molecular rearrangements, mechanistic pathways of substitution.

INSTRUCTIONS FOR PAPER SETTERS AND CANDIDATES :

(i) *Examiner will set total of **Nine** questions comprising **Two** questions from each unit and **One** compulsory question of short answer type covering whole syllabi.*

(ii) *The students are required to attempt **Five** questions in all, **One** question from each unit and the Compulsory question.*

(iii) *All questions carry equal marks.*

Suggested Books:

1. Cotton, F.A., G. Wilkinson, *Advanced Inorganic Chemistry;* 6th edition Pubs: John Wiley & Sons, 1999.

2. Basolo, F., Pearson, R.G., *Inorganic Reactions;* Reprint 2nd edition, Pubs: Wiley-Eastern, 1977.

3. Greenwood, N.N., Earnshaw, A., *Chemistry of Elements;* 1st edition, Pubs: Pergamon Press, 1984.

4. Huyer, J.E., Keiter, E.A., Keiter, R.L., *Inorganic Chemistry-Principles of Structure and Reactivity;* 4th edition, Pubs: Pearson Education India, 1993.

Paper IV: PHYSICAL CHEMISTRY

Total Marks	:	100
Theory	:	90
Internal Assessment	:	10
Total Lectures	:	40

OBJECTIVE OF THE COURSE :

To teach the fundamental concept of Chemistry and their applications, the syllabus pertaining to B.Sc. (Honours) in the subject of Chemistry has been upgraded as per provision of the UGC module and demand of the academic environment. The course contents have been revised from time to time as per suggestions of the teachers of the Chemistry working in the Panjab University, Chandigarh and affiliated colleges. The syllabus contents are duly arranged unit wise and contents are included in such a manner so that due importance is given to requisite intellectual and laboratory skills.

UNIT-I

1. **Transport Properties :** (10 Lectures)

Recapitulation of Collision Properties of gases such as collision cross section, collision diameter, collision number, collision frequency and mean free path.

Meaning of transport, Properties of gases, Flux, General equation of transport. Thermal conductivity, Kinetic theory of thermal conductivity (derivation of the expression).

Viscosity of gases, kinetic theory of gas viscosity (derivation of the expression). Measurement of gas viscosity, Diffusion Coefficient. Kinetic theory of diffusion in gases (derivation of the expression).

Diffusion in liquids, theory of diffusion in liquids, diffusion in solutions, laws of diffusion.

UNIT-II

2. **Dynamic Electro-Chemistry :** (5 Lectures)

Processes at electrodes. The electric double layer (Gouy-Chapman, Helmholtz and Stern models–qualitative ideas. The potential difference, Rate of charge transfer, Bulter Volmer equation, Tafel plot, Overpotential Polarisation, Polarography, Fuel cells, Corrosion, Inhibition of Corrosion.

Surface Chemistry : (5 Lectures)

Interfacial region, types of interfaces, surface tension, thermodynamics of surfaces, plane interface, curved surfaces, Laplace and Kelvin equations including their derivation.

Gibbs Adsorption Equation and its Thermodynamic derivation, Wetting and adhesion in solid-liquid systems, contact angle, Capillary rise and surface tension. Surface Films on liquids.

UNIT-III

3. **Colloids :** (10 Lectures)

The colloidal state; preparation of colloidal solutions, Methods of purification of colloidal systems, dialysis, electro-dialysis, filtration, Ultra-filtration. Classification of colloidal systems, their typical properties with special reference to optical properties (Tyndall effect). The Brownian movement. Electrokinetic phenomena, Stabilization of colloidal systems and theories of stability; zeta potential, Coagulation, Flocculation of colloids by electrolytes and its mechanism. Association Colloids, Micellar system, Emulsions and their types, HLB value of emulsifiers, Introduction to microemulsions.

UNIT-IV

4. **Quantum Chemistry :** (10 Lectures)

ATOMS : Schrodinger equation for multi-electron atoms, Need for approximate Solutions. Basic ideas of the Variation theorem and the Perturbation methods. Self-Consistent Field methods (Hertree and Hertree-Fock) - Elementary treatment only. Indistinguishability of Electrons. Pauli exclusion principle, two electron spin functions; Slater determinants and Pauli principle. New Atomic term symbols. Spin-orbit coupling (Total orbital momentum, multiplicity, total angular momentum) vector model of atom, Hund's rules, atomic spectra and Selection rules.

Molecules :

Schrodinger equation for molecules, Born-Oppenheimer approximation.Valence bond theory and Molecular Orbital theory as two approaches for approximate solution of Schrodinger equation LCAO-MO treatments of H_2 and Valence bond treatment of H_2 (complete solutions not required. Only wave functions to be discussed) Comparison of MO and VB treatments. Brief Introduction to the electronic structure for polyatomic molecules. Metals, Insulators and Semiconductors. Band Theory of Solids (elementary discussion).

INSTRUCTIONS FOR PAPER SETTERS AND CANDIDATES :

(i) *Examiner will set total of **Nine** questions comprising **Two** questions from each unit and **One** compulsory question of short answer type covering whole syllabi.*

(ii) *The students are required to attempt **Five** questions in all, **One** question from each unit and the Compulsory question.*

(iii) *All questions carry equal marks.*

Suggested Books :

1. Atkins, P.W., *Physical Chemistry;* 8th edition, Pubs: Oxford University Press, 2008.

2. Levine, I.N., *Physical Chemistry,* 3rd edition, Pubs: Prentice Hall International Inc., 2000.

3. Pruton, Maron, *Principles of Physical Chemistry;* 4th edition, Pubs: Oxford & IBH Publishing Co. Pvt. Ltd., 1972.

4. Chandra, A.K., *Introductory Quantum Chemistry;* 4th edition, Pubs: Tata McGraw-Hill Publishing Company Ltd.

5. Moore, W.J., *Basic Physical Chemistry;* 1st edition, Pubs: Prentice-Hall of India Pvt. Ltd., 1986.

6. Albert, R.A., *Physical Chemistry;* 1st edition, Pubs: John-Wiley & Sons, 1992.

BOTANY

B.Sc. (HONOURS) THIRD YEAR EXAMINATION, 2011

		Theory	*Int. Assess.*	*Total*	*Practical*	*Int. Assess.*	*Total*
Theory Paper-A :	Biology of Cryptogams, Seed Plants, Microbiology and Plant Pathology	68	+ 7	75	22.5	+ 2.5	25
Theory Paper-B :	Genetics and Plant Breeding, Biochemistry, Molecular Biology and Biotechnology, Systematic of Angiosperms and Environmental Biology	68	+ 7	75	22.5	+ 2.5	25
	Total :			**150**			**50**

Note : 1. The number of teaching hours for theory and practical per session shall be 120 hrs. and 200 hrs. respectively.

2. There will be a total of nine questions in each theory papers A & B. Question No. 1 will be compulsory and will consist of 20 parts (one mark each) comprising 10 MCQ and the rest 10 parts will be of fill-in the blanks covering the entire syllabus in both the theory papers A & B. The remaining 8 questions in A & B shall include two questions from each unit. Candidates shall be required to attempt one question from each Unit. Question No. 1 will carry 20 marks and the rest of 8 questions will be of 12 marks each.

PAPER-I : BIOLOGY OF CRYPTOGAMS; SEED PLANTS, MICROBIOLOGY & PLANT PATHOLOGY

Max. Marks : 75
Theory : 68
Internal Assessment : 07

Objective : The basic objective of this paper is to make the students understand the diversity in various life forms of plant kingdom so as to have knowledge about evolution of simpler forms to complex ones. Simultaneously, it also aims to make students familiar with the systematic position of micro-organisms, their ultrastructure, classification and industrial application. It also deals with the important aspect of plant diseases, their pathogens and disease managements.

Teaching Methodology : Teaching methodology includes series of lectures, making use of charts, transparencies, LCD, Models, slides, practical demonstrations, extension lectures from experts, field visits, discussions, quiz competitions etc. In practicals, students would be provided with fresh/preserved materials for their morphological and anatomical studies making use of microscopes and binoculars and hands-on tools/equipment etc.

UNIT-I

BIOLOGY OF CRYPTOGAMS

Algae : Occurrence and distribution; thallus organization and evolutionary tendencies; ultrastructure of algal cell; criteria for classification and typical life histories of algae belonging to various classes; role of algae in human welfare.

Fungi : Occurrence and distribution; ultrastructure of fungal cell, cell wall composition, modern concepts in the classification of fungi; nutrition, reproduction, life histories of main groups of fungi; types of fungal spores and mode of their liberation; role of fungi in human welfare.

Bryophytes : Comparative account of morphology, anatomy and reproduction in *Riccia, Marchantia*, *Anthoceros*, *Funaria*; evolution of sporophyte and gametophyte; classification, ecological aspects and economic importance.

Pteridophytes : The primary vascular plants; classification, comparative account of morphology, anatomy and reproduction in *Rhynia, Lycopodium, Selaginella, Equisetum, Marsilea, Pteris*, Telome theory; Stelar evolution; heterospory and seed habit.

Suggested Readings :

1. Bold, H.C., Alexopoulos, C.J. and Delevoryas, T., *Morphology of Plant and Fungi* (4th Edition), Harper and Foul Co., New York, 1980.

2. Dube, H.C., *An Introduction to Fungi,* Vikas Publishing House Pvt. Ltd., Delhi, 1990.

3. Glifford, E.M. and Foster, A.S., *Morphology and Evolution of Vascular Plants,* W.H. Freeman & Co., New York, 1989.

4. Glibert, M.S.*, Cryptogamic Botany,* Vol. I & II (2nd Edition), Tata McGraw Hill Publishing Co. Ltd., New Delhi, 1985.

5. Kumar, H.D., *Introductory Phycology,* Affiliated East-West Press Ltd., New York, 1988.

6. Puri, P., *Brophytes,* Atmaram & Sons, Delhi, 1985.

7. Rangaswamy, G. and Mahadevan, A., *Diseases of Crop Plants in India,* Prentice Hall India Pvt. Ltd., New Delhi, 1999.

8. Wilson, N.S. and Rothwell, G.W., *Palaeobotany and the Evolution of Plants* (2nd Edition). Cambridge University Press, U.K., 1993.

UNIT-II

SEED PLANTS :

Phanerogams, the seed bearing plants : General characteristics and types.

Gymnosperms : General characters, ontogeny and structure of seed; the ovule and megasporogenesis, female gametophyte; pollination, formation of pollen tubes and fertilization; embryogeny and maturation of seed.

The Living Cycads : Distribution; vegetative organography and anatomy; reproductive cycle—sporophytes and sporangia, gametophytes, fertilization, embryogeny and seed development.

The Coniferales : General organography and anatomy; foliage leaves; storbilli and sporangia; reproductive cycle-sporogenesis, gametophytes, fertilization, embryogeny and seed development.

The Gnetales : Habit and distribution; vegetative organography and anatomy; reproductive cycle –the strobilli, sporogenesis, male and female gametophytes, pollination, fertilization and embryogeny.

Angiosperms : Organography and anatomy; leaf morphology, histology and venation; nodal and wood anatomy; general concepts of floral anatomy.

Alternatives for Reproduction : Vegetative and sexual reproduction; transition from vegetative to reproductive development.

Flower : Evolution, concept of flower as a modified determinate shoot; genetic control of floral organs; functions of flower.

Structure of Anther : Microsporogenesis : Formation of pollen grains (male gametophyte); pollen germination; pollen tube growth.

Structure of Pistil : Ovules; megasporogenesis; development of embryo sac (female gametophyte).

Mechanisms and Agencies of Pollination : Pollen-stigma interaction; self-incompatibility, double fertilization, apomixis.

Seed and Fruit : Development of endosperm and embryo in monocotyledons and dicotyledons; storage of reserve materials and desiccation in seeds; dormancy and seed germination; fruit maturation; ripening and dispersal.

Suggested Readings :

1. Cronquist, A., *The Evolution and Classification of Flowering Plants,* Thomas Nelson (Printers) Ltd., London and Edinburgh, 1968.

2. Delevoryas, Th. *Plant Diversification, Modern Biology Series,* Halt, Rinehart and Wintson, New York, 1965.

3. Foster, A.S. and Gifford, A.E.M., Jr., *The Comparative Morphology of Vascular Platns,* Vakils, Peffer and Simons Pvt. Ltd., 1967.

4. Sporne, K.R., *The Morphology of Angiosperms,* B.I. Publications, Bombay, 1977.

5. Bhojwani, S.S. and Bhatnagar, S.P., *The Embryology of Angiosperms,* 4th revised and enlarged edition, Vikas Publishing House, Delhi, 2000.

6. Johri, B.M., *Embryology of Angiosperms,* Springer-Verlag, Berlin, 1984.

7. Raghvan, V., *Molecular Embryology of Flowering Plants,* Cambridge University Press, New York, 1997.

UNIT-III

MICROBIOLOGY :

Discovery of Microorganisms : Systematic position of microorganisms in biological world; classification of microorganisms and characteristic features of different groups.

Methods of Microbiology : Basic principles of microscopy, micrometry, staining sterilization methods, culture media, pure culture methods, methods of population estimation, growth determination.

Ultrastructure of Microorganisms : Prokaryotic microorganisms; fine structure of prokaryotic cell; eukaryotic microorganisms; viruses—properties and classification; characteristic features of host-virus interaction; bacteriophage T4; tobacco mosaic virus; general account of mycoplasma and actinomycetes.

Genetic Recombination in Prokaryotes : Conjugation, transformation and transduction.

Role of Microorganisms in Biogeochemical cycling of nitrogen and carbon; Biological N_2 fixation.

Industrial Application of Microorganisms : Organic acids, alcohol, food processing, milk products, antibiotics, biopesticides.

UNIT-IV

PLANT PATHOLOGY :

General Account of Plant Pathogens : Historical developments; general account of diseases caused by plant pathogens.

Major Diseases of Plants and their Control : Blast and brown spot of rice, rust and smuts of wheat, downy mildew and green ear disease of bajra, white rust of crucifers, late blight of potato, wilt of pigeon pea, damping off seedlings of tomato/mustard/chillies, Tikka or leaf spot of groundnut, citrus die back, powdery mildews of cucurbits and grapes, red rot of sugarcane.

Pathogen Attack and Defense Mechanisms : Physical, physiological, biochemical and molecular aspects.

Plant Disease Epidemology : Transmission and spread of plant pathogens; disease cycles, epidemics; modelling and disease forecasting.

Plant Disease Management : Chemical, biological; IPM systems; development of transgenics; biopesticides; plant disease clinics.

Genetics of Resistance and Susceptibility : Genes for virulence and avirulence, their application in resistance and susceptibility: induced resistance (immunization).

Molecular Plant Pathology : Molecular diagnosis; identification of genes and specific molecules in disease development; molecular manipulation of resistance.

Application of Information Technology in Plant Pathology : Simulation of epidemics; programmes for diagnosis; remote sensing and image analysis for ecosystem level effects; prediction of disease control decisions.

Suggested Readings :

1. Agrios, G.N., *Plant Pathology,* Academic Press, London, 1997.
2. Albajes, R. Gullino, M.L., Van Lenteren, J.C. and Elad, Y., *Integrated Pest and Disease Management in Green House Crops,* Kluwer Academic Publishers, 2000.
3. Bridge, P. et al., *Molecular Variability of Fungal Pathogens,* CAB International, UK, 1998.
4. Bridge, P. et al., *Application of PCR in Mycology,* CAB International, UK, 1999.
5. Bridge, P., Moore, D.R. and Scott, P.R., *Informational Technology, Plant Pathology and Biodiversity,* CAB International, UK, 1998.
6. Clifton, A., *Introduction to the Bacteria,* McGraw Hill Co., New York, 1958.
7. Mandahar, C.L., *Introduction to Plant Viruses,* Chand & Co. Ltd., Delhi, 1998.
8. Persley, G.J., *Biotechnologies and Integrated Pest Management,* CAB International, UK, 1996.
9. Skerritt, J.H. and Apples, R., *New Diagnostics in Crop Sciences,* CAB International, UK, 1995.

Teachers may collect plants/material available in their locality/institution.

Max. Marks	:	25
Practical	:	22.5
Int. assessment	:	2.5

Suggested Laboratory Exercises (Biology of Cryptogams) :

1. Comparative study of algal genera with reference to the classes mentioned in theory.

2. Comparative study of fungal genera with reference to the classes mentioned in theory.

3. Fungal diseases: White rust of crucifers, downy mildew of pearl millet, green ear disease of bajra, powdery mildew, rust of wheat and sorghum, Tikka or leaf spot disease of ground nut, red rot of sugar cane.

4. Section cutting of fungal diseased specimens and preparation of lactophenol mounts.

5. Comparative study of morphology, anatomy and reproductive structures in *Marchantia, Anthoceros* and *Funaria.*

Suggested Laboratory Exercises (Seed Plants) :

1. Comparative anatomical structure of stem/wood of Cycadales, Coniferales, Ginkgoales, Ephedrales, Gnetales and angiosperms, from sections and macerations.

2. Comparative structure of leaf, leaf appendages, venation and stomata of gymnosperms and angiosperms.

3. Comparative structure of the male and female cones of Cycads, Conifers, Ginkgoales (Wherever available) and Ephedrales and flowers of Ranales and Magnoliales.

4. Diversity in pollen grain structure among Cycads, Conifers, Ephedrales and angiosperms with respect to size, exine sculpturing, germpore number, number of constituent cells and any other features.

5. Microsporogenesis and male gametophyte development in angiosperms through microtome sectioning and acetocarmine squash technique.

6. Pollen grain germination by hanging-drop and sitting-drop techniques in *Impatiens, Catharanthus roseus* (= Vinca rosea) or any other suitable material.

7. Diversity in the structure of stigma, style, stigmatic papillae and transmitting tissue of style.

8. Comparison of mating types in species exhibiting heteromorphic self-incompatibility *Hamelia patens, Pentas* or any other suitable material.

9. Studies on gametophytic and sporophytic self-incompatibility in *Petunia.*

10. Microdissection of embryo with suspensor at different stages of development to unravel relationship between the two.

11. Microdissection of endosperm with different types of haustoria.

Suggested Laboratory Exercises (Microbiology) :

1. Calibration of microscope: determination of dimensions of microorganisms (suggested model organisms: yeast, lactobacilli, cyanobacteria).

2. Cultivation media for autotrophic and heterotrophic microorganisms (cleaning of glasswares, mineral media, complex media, solid media, sterilization) (based on topic 3).

3. Isolation of microorganisms : Streaking on agar plates/pour plate method, isolation of clones, preservation (based on topic 2 and 3).

4. Determination of growth of a microorganism (model organism : *Escherichia coli,* effects of nutrients, e.g. glucose, fructose, sucrose; principle of colorimetry/spectrocolorimeter) (based on topic 3).

5. Determination of microbial population size (suggested model organism: yeast; use of haemocytomter, serial dilution technique, relationship between dilution and cell count, determination of standard error, reliability in cell counts) (based on topic 3).

6. Preparation of Winogradsky column using pond bottom mud, observations on temporal sequence of appearance of microbes. (visual appearance, microscopic observations) (based on topic 7).

7. Observation on virus infected plants (symptoms) (based on topic 5).

8. Fermentation by yeast (inverted tube method, use of different substrates, e.g. glucose, fructose, cane sugar, starch) (based on topic 8).

Suggested Laboratory Exercises (Plant Pathology) :

1. Study of important plant pathogens (symptoms and host parasite relationship).

2. Isolation and culture of plant pathogens (e.g. *Colletotrichum, Fusarium, Alternaria*) and establishment of Koch's postulates and their pathogenicity.

3. Effect of pathogens on physiology of a host.

4. Isolation of cellulose from diseased plants.

5. Isolation of pectolytic enzymes from diseased plants.

6. Study on antagonism between isolated antagonists and plant pathogens and test of biological control.

7. Demonstration of antibiosis using a bacterial culture and known antibiotics.

8. Demonstration of the assay of prohibitins (phytoalexins).

9. Demonstration of biopesticides (essential oils, neem, turmeric and garlic) against some pathogens.

Paper-B : GENETICS AND PLANT BREEDING; BIOCHEMISTRY MOLECULAR BIOLOGY AND BIOTECHNOLOGY, SYSTEMATIC OF ANGIOSPERMS AND ENVIRONMENTAL BIOLOGY

Max. Marks	:	75
Theory	:	68
Internal Assessment	:	07

Note : 1. The number of teaching hours for theory and practical per session shall be 120 hrs. and 200 hrs. respectively.

2. There will be a total of nine questions in each theory papers A & B. Question No. 1 will be compulsory and will consist of 20 parts (one mark each) comprising 10 MCQ and the rest 10 parts will be of fill-in the blanks covering the entire syllabus in both the theory papers A & B. The remaining 8 questions in papers A & B shall include two questions from each unit. Candidates shall be required to attempt one question from each Unit. Question No. 1 will carry 20 marks and the rest of 8 questions will be of 12 marks each.

Objective : This paper deals with taxonomy and species concept which makes the basis for further advanced study in Botany as without having a knowledge of identification, nomenclature and classification no research work can be pursued. Besides a knowledge of molecular and cellular basis would enable the students to understand the hereditary and evolutionary trends in the plant Kingdom. This paper also provides a brief knowledge about inter-relations of plants and environment as well as various environmental problems like Global Warming, ozone depletion, thermal and radioactive pollution, prevention and control of pollution along with role of national and international organizations in environmental management.

Teaching Methodology : Teaching methodology includes series of lectures, making use of charts, transparencies, LCD, Models, slides, practical demonstrations, extension lectures from experts, field visits, discussions, quiz competitions etc. In practicals, students would be provided with fresh/preserved materials for their morphological and anatomical studies making use of microscopes and binoculars and hands-on tools/equipment etc.

UNIT-I

GENETICS :

Mendel's Experiments and Principles of Inheritance : Backcross and test cross; gene interactions and modified dihybrid ratios—Complementary, supplementary, duplicate and epistatic factors.

Multiple Allelism : Multiple alleles in *Drosophila* (eye colour), man (blood groups) and plants (self incompatibility).

Quantitative Genetics : Quantitative traits and quantitative genetics; the multiple factor hypothesis; descriptive statistics.

Linkage and Recombination : Coupling and repulsion phases; two and three point test crosses with their significance in chromosome mapping ; interference and co-efficient of coincidence.

Sex Chromosomes in *Drosophila,* Man and *Melandrium* : Balance concept of sex determination in *Drosophila*; mechanisms of sex determination; sex-linked inheritance in *Drosophila* and man; sex limited characters.

Maternal Influence on Inheritance : Shell coiling in snails and Kappa particles in *Paramecium*; cytoplasmic inheritance in yeast (mitochondria) and *Mirabilis Jalapa* (plastids).

Alterations in the Genetic Make up – changes at Genetic Level : Spontaneous and induced mutations; mutagens–types and mode of action; transitions, transversions and frame-shift mutations; detection of mutations.

Alterations in Genetic Make up – changes in Chromosome Structure : Origin, types and effects of duplications, deletions, inversions and translocations; meiosis in structural heterozygotes.

Alterations in Genetic make up – changes in Chromosome Number : Origin, types and effects of auto and allopolyploidy; origin and meiosis in nullisomics, monosomics and trisomics.

PLANT BREEDING :

Types of Plant Reproduction : Vegetative, sexual and apomixis; their effects on generating and fixing genotypic variation.

Methods of Plant Improvement : Pure line and mass selection; hybridization in self and cross pollinated crops; introduction and acclimatization; hybrid vigour.

Mutations and polyploidy as methods of plant improvement.

Suggested Readings :

1. Atherly, A.G., Girton, J.R. and McDonald, *The Science of Genetics,* Saunders College Publishing Co., Fort Worth, USA, 1999.

2. Gardener, J., Simmons, H.J. and Snustad, D.P., *Principles of Genetics* (8th Edition), John Wiley & Sons, New York, 1991.

3. Gupta, P.K., *Genetics,* Rastogi Publications, Shivaji Road, Meerut, 1994.

4. Gupta, P.K., *Cytogenetics,* Rastogi Publications, Shivaji Road, Meerut, 1995.

5. Hartl, D.L. and Jones, E.W., *Genetics: Principles and Analysis* (4th Edition), Jones & Batlett Publishers,. Massachusetts, USA, 1998.

6. Poehlmann, J.M. and Sleeper, D.R., *Breeding Field Crops,* Panima Publishing House, New Delhi, 1995.

7. Russel, P.J., *Genetics* (5th Edition), The Benjamin/Cummings Publishing Co., Inc., USA, 1998.

8. Simmonds, N.W., *Principles of Crop Improvement,* Longman, London and New York, 1979.

9. Snustad, D.P. and Simmons, M.J., *Principles of Genetics* (2nd Edition), John Wiley & Sons, Inc., USA, 2000.

10. Sharma, J.R., *Principles and Practice of Plant Breeding,* Tata McGraw Hill Publishing Co. Ltd., New Delhi, 1994.

UNIT-II

BIOCHEMISTRY, MOLECULAR BIOLOGY AND BIOTECHNOLOGY :

Nucleic Acids : Composition of nucleic acids and synthesis of nucleotides; DNA structure; A, B and Z forms of DNA; denaturation and renaturation of DNA; chromatin structure; DNA replication and recombination; DNA polymerases; different forms of RNA and their role.

Amino Acid and Protein Metabolism : Structure, characteristics and classification of amino acids; protein and non-protein amino acids; amino acid biosynthesis; GS/GOGAT cycle; transamination; peptide bond and polypeptide chain; primary, secondary, tertiary and quaternary structure of proteins; protein biosynthesis and its regulation; post-translational modification of proteins; protein targeting; protein degradation.

Carbohydrate Metabolism : Classification; structure of some representatives examples of monosaccharides, disaccharides, polysaccharides; stereoisomers, enantiomers and epimers; biosynthesis and degradation of sucrose and starch.

Lipid Metabolism : Saturated and unsaturated fatty acids; fatty acid biosynthesis; oxidation of fatty acids; storage and mobilization of fatty acids and lipids.

Gene Structure, Expression and Regulation : Gene organization in prokaryotes and eukaryotes; operon concept; gene regulation in prokaryotes and eukaryotes; inducible, repressible, positive and negative gene regulation; interrupted genes in eukaryotes; RNA splicing; mRNA stability.

Recombinant DNA Technology : Restriction endonucleases : Prokaryotic and eukaryotic cloning vectors; genomic and cDNA libraries; southern and northern analysis; various techniques of gene mapping and DNA fingerprinting (RFLP, RAPD, AFLP); chromosome walking; polymerase chain reaction; DNA sequencing.

Plant Biotechnology : Cellular differentiation and totipotency; organogenesis and embryogenesis; protoplast isolation and culture; somatic hybridization; clonal propagation; genetic engineering of plants; vectors for gene delivery; selectable markers and reporter genes; methods of gene delivery; agrobacterium –the natural genetic engineer; salient achievements in crop biotechnology (with suitable examples) and prospects.

Suggested Readings :

1. Alberts, B., Bray, D., Lewis, J., Raff, M., Roberts, K. and Watson, J.D., *Molecular Biology of Cell,* Garland Publishing Co. Inc., New York, USA, 1999.

2. Bhojwani, S.S., *Plant Tissue Culture: Applications and Limitations,* Elsevier Science Publishers, New York, USA, 1990.

3. Buchanan, B.B., Gruissem, W. and Jones, R.L., *Biochemistry and Molecular Biology of Plants,* American Society of Plant Physiologists, Maryland, USA, 2000.

4. Collins, H.A., and Edwards, S., *Plants Cell Culture,* Bios Scientific Publishers, Oxford, UK, 1998.

5. Dennis, D.T., Turpin, D.H., Lefebvre, D.D. and Layzell, D.B. (eds.), *Plant Metabolism* (2nd Edition). Longman, Essex, England, 1997.

6. Lea, P.J. and Leegood, R.C., *Plant Biochemistry and Molecular Biology* (2nd Edition), John Wiley and Sons, Chichester, England, 1999.

7. Lodish, H., Berk, A., Zipursky, S.L., Matsudaira, P., Baltimore, D. and Darnell, J., *Molecular Cell Biology* (4th Edition), W.H. Freeman and Co., New York, U.S.A., 2000.

8. Old, R.W. and Primrose, S.B., *Principles of Gene Manipulation,* Blackwell Scientific Publications, Oxford, UK, 1989.

9. Raghavan, V., *Embryogenesis in Angiosperms : A Developmental and Experimental Study,* Cambridge University Press, New York, USA, 1986.

10. Vasil, I.K. and Thorpe, T.A., *Plant Cell and Tissue Culture,* Kluwer Academic Publishers, the Netherlands, 1994.

UNIT-III

ANGIOSPERMS SYSTEMATICS :

Introduction : Aims and components of systematics, introduction to identification, nomenclature, phylogeny and classification.

Systematics in Practice : Importance of herbarium specimens and their preparation; role of herbaria and botanical gardens; documentation (floras, monographs, manuals, journals, abstracts, indices and dictionaries); keys for identification of plants—single access and multi access; value of computers and databases for identification.

Taxonomic Hierarchy : Taxonomic category, taxonomic groups; concepts of species, genus and family.

Botanical Nomenclature : Principles and rules; ranks and names; type method; principle of priority and its limitations; names of hybrids and cultivars; concept of biocode.

Phylogeny of Angiosperms : A general account of the origin and evolution of angiosperms (special reference to Bennettitalean, Gnetalean, Caytonialean and herbaceous origin theories); primitive living angiosperms.

Systems of Classification : Bentham and Hooker's system. Engler and Prantl's system and Takhtajan's system.

Modern Taxonomy : Supporting evidences/inputs for taxonomy; taxonomy in relation to anatomy, embryology, palynology, ecology, cytology (cytotaxonomy), secondary metabolites in plants (chematoxonomy).

Numerical Taxonomy : Concepts, characters and attributes; OTU's coding; cluster analysis; cladistics.

UNIT-IV

ENVIRONMENTAL BIOLOGY :

Introduction : Inter-relationship between the living world and the environment; the components and dynamism; homeostasis; relevance to man.

Earth as a System : The biosphere, the hydrosphere, the atmosphere and the lithosphere; components within biosphere (Biomes); parameters delimiting individual biomes.

The Environment : Soil—general account and adaptations; water—general account and adaptations; the atmosphere—general account and adaptations; the living world—biotic component of environment; types of biotic interactions; fire as an ecological factor.

Organismal Ecology/Biotic Components : Individuals, species, population communities and their characteristics.

Ecosystems : Concepts of ecosystem; homeostasis; structure of ecosystem; functions of ecosystem; transfer of energy and minerals via grazing and detritus chains and role of micro-organisms; cycles (hydrologic, gaseous); role of humans in maintaining biogeochemical cycles.

Diversity of Ecosystem : Aquatic (fresh water); terrestrial (forest/grassland); man-made ecosystems.

Phytogeography : Introduction; endemism, static and dynamic plant geography; a short account of vegetation of India.

Human Ecology and Ecological Management : The human population; renewable and non-renewable natural resources and their management; conservation of biodiversity; endangered species; conventional and non-conventional energy sources.

Impact of Human Activities : Pollution of air, water and soil; a brief account of environmental toxicology; incidence of noise; thermal and radioactive pollution; prevention and control of pollution; global warming, desertification and ozone depletion.

Role of National and International Organizations in Environmental Management : Formulation of optimal models.

Bio-indicators.

Environmental Impact Assessment : A brief account.

Teachers may collect plants/material available in their locality/institution.

Max. Marks	:	25
Practical	:	22.5
Int. assessment	:	2.5

Suggested Laboratory Exercises (Genetics and Plant Breeding) :

1. Determination of chromosome count from dividing pollen mother cells, root tips and pollen grains.
2. Preparation of karyotypes from diving root tip cells and pollen grains.
3. Determination of instraspecific variation in chromosome number from locally available taxa.
4. Study of sex chromosomes and their behaviour during meiosis from grasshopper and any appropriate dioecious plant (e.g. *Coccinia*).
5. Detection of anomalies in chromosome pairing and disjunction caused by mutant genes and structural alterations of chromosomes.
6. Preparation of chromosome maps from 3-point test cross data.
7. Identification of mutant phenotypes in *Drosophila* and *Arabidospsis* stocks maintained by the Department.
8. Correlation of floral structure with pollination system (e.g. *Salvia, Sesamum,* pea, *Lathyrus,* wheat, rice, maize).
9. Field exploration for detection of male sterile plants and estimation of their pollen fertility in locally grown crop plants e.g. *Sorghum,* tomato and *Linum.*
10. Estimation of pollen ovule ratio and its bearing on pollination system.
11. Emasculation and bagging of flowers of Brassicaceae, Poaceae, Paplionaceae, Malvaceae and Linaceae, pollinating them manually and estimating fruit and seed set.

Suggested Laboratory Exercises (Biochemistry, Molecular Biology and Biotechnology) :

1. Chemical tests to demonstrate the presence of starch, sugar, fat and protein in plant material.
2. To identify the amino acids in a mixture by resolving through paper chromatography or TLC.
3. To prepare the standard curve for protein and determine the protein content in unknown samples by Biuret method.
4. Colorimetric estimation of RNA using orcinol.
5. Colorimetric estimation of DNA using diphenyl amine.
6. Isolation of plant genomic DNA and its spooling.
7. Isolation of total RNA from plant tissue and its colorimetric estimation.
8. Preparation of tissue culture media, sterilization and inoculation of plant material.
9. Demonstration of techniques of in vitro culture of various explants.
10. Isolation of plant protoplasts (e.g. Tobacco, *Petunia*) using enzymes available commercially and estimation of their yield.

Note : In the practical classes, emphasis should be given on basic principles of spectro-photometry, chromatography, electrophoresis and rDNA technology and related fundamentals.

Suggested Laboratory Exercises (Systematics of Angiosperms) :

Description of the locally available species of the following families & genera :

1. Ranunculaceae : *Ranunculus, Delphinium.*
2. Brassicaceae : *Brassica, Iberis.*
3. Capparidaceae : Capparis/*Cleome.*
4. Caryophyllaceae : *Dianthus, Stellaria.*
5. Rutaceae : *Citrus, Murraya*
6. Tiliaceae : *Corchorus.*
7. Fabaceae : Faboideae : *Lathyrus, Melilotus,* Caesalpinioideae : *Cassia, Caesalpinia;* Mimosoideae : *Mimosa, Acacia.*

8. Myrtaceae : *Callistemon, Eucalyptus.*
9. Cucurbitaceae : *Luffa.*
10. Apiaceae : *Coriandrum.*
11. Rubiaceae : *Hamelia/Mussaenda.*
12. Asteraceae : *Helianthus, Ageratum, Sonchus.*
13. Apocyanaceae : *Vinca, Nerium, Tabernaemontana.*
14. Asclepiadaceae : *Calotropis.*
15. Solanaceae : *Solanum, Petunia.*
16. Acanthaceae : *Adhatoda, Peristrophe.*
17. Lamiaceae : *Ocimum, Salvia.*
18. Chenopodiaceae : *Chenopodium.*
19. Euphorbiaceae : *Euphorbia, Phyllanthus.*
20. Moraceae : *Morus, Ficus.*
21. Cannaceae : *Canna.*
22. Liliaceae : *Asphodelus, Asparagus.*
23. Commelianaceae : *Tradescantia, Commelina.*
24. Poaceae : *Avena, Triticum.*

Suggested Laboratory Exercises (Environmental Biology) :

1. Mechanical analysis of soils by sieve method.
2. Determination of soil porosity and density (sand and pit method).
3. Determination of water holding capacity and field capacity of soil.
4. Determination of permeability (capillarity and percolation) of different types of soils.
5. Titrimetric estimation of total carbonates of soil samples.

6. Quantitative determination of soil organic matter by Walklay and Black's rapid titration method.

7. Determination of species area curve by minimal quadrat size.

8. Analysis of the herbaceous vegetation for frequency, density and abundance.

9. Study the height spectrum of herbaceous vegetation by line transect method.

10. Effect of tree canopy on the distribution of herbaceous vegetation.

11. Estimation of biomass of aerial parts of herbaceous plants (fresh weight and dry weight).

12. Analysis of different water samples for pH, oxygen, carbon-dioxide (titrimetric estimation), turbidity and temperature.

13. Demonstration of desert and aquatic ecosystems with the help of models.

14. **Field Visit :** students should be taken for field visits to places of ecological/environmental interest. They should submit detailed report of the visit in the form of project report in the final practical examination for evaluation. The report shall carry marks.

Note to teachers :

The students are to be familiarized with the families listed above in the practical classes with representatives species or any other that may be available locally. However, questions pertaining to these may be asked in the theory examinations.

ZOOLOGY

B. Sc. (HONOURS) THIRD YEAR EXAMINATION, 2011

Paper-III : REPRODUCTIVE BIOLOGY AND HUMAN WELFARE

Max. Marks	:	75
Theory	:	67 marks
Internal Assessment	:	8 marks
Time	:	3 Hrs.

Note : Nine questions are to be set. Question No. 1 is compulsory consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1½ marks each. Two questions are to be set from each unit. One question is to be attempted from each unit. In all, Five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts.

UNIT-I

1. *Physiology of Male Reproductive System :*
   (a) Structure and functions of testis, epididymis, vas deferens, seminal vesicle and prostate gland.
   (b) Semen composition, sperm maturation and capacitation.

UNIT-II

2. *Physiology of Female Reproductive System :*
   (a) Structure and functions of ovary, oviduct, uterus and vagina.
   (b) Oestrous and menstural cycles.
   (c) Fertilization, implantation, gestation, parturition and lactation.

UNIT-III

3. *Neuro-Hormonal Regulation of Production :*
   (a) Gonadal hormones.
   (b) Gonadotropic hormones.
   (c) Hypothalamic factors.
   (d) Structural behaviour.

UNIT-IV

4. *Fertility Regulation :*

   (a) Contraception and modern contraceptive technology – natural, mechanical, chemical immunological and surgical methods and Fertilization *in vitro* (Test tube baby).

   (b) Application of fertility regulation in live stock :

      (i) Artificial insemination in farm animals.

      (ii) Milk yield.

      (iii) Egg production in poultry.

Books Recommended :

1. Milt, J. and Everitt, B.J., *Essentials of Reproduction,* Blackwell Scientific Publications,1985.

2. Genong, M., *Reproduction,* Saunders Intern. Publications, 1985.

3. Turner, C.P. and Bagnara, J.T., *General Endocronology,* Saunders Intern. Publications, 1984.

4. Greep, R.O. and Astwood, E.B. (Eds.), *Handbook of Physiology,* Kluwer Academic/Plenum Publishers, New York, 2002.

5. Balin, H. and Glasser, S. (Eds.), *Reproductive Biology,* Excerpta Media, 1972.

6. Guyton, A.S., *Text Book of Medical Physiology,* 7th Edition, Iquakushoinl/Saunders International, Publications, 1994.

Paper-IV : AQUATIC BIOLOGY

Max. Marks	:	75
Theory	:	67 marks
Internal Assessment	:	8 marks
Time	:	3 Hrs.

Note : Nine questions are to be set. Question No. 1 is compulsory consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1½ marks each. Two questions are to be set from each unit. One question is to be attempted from each unit. In all, Five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts.

UNIT-I

1. *Ecosystem :* Study of ecosystem with particular reference to the components of aquatic ecosystem. Food chains and food webs. Flow of energy in a food chain.

2. *Physico-chemical characteristics of Freshwater Environment :* Temperature, light, current of water, density, pH, dissolved oxygen, carbon dioxide and nutrients.

UNIT-II

3. *Freshwater zonation in a lake and stream :* Supralittoral, littoral, sublittoral, profundal & limnetic zones of a lake. Pool zone and rapid zone of a stream.

4. *Classification of freshwater habitats :* Streams, ponds, lakes, marshes and swamps.

UNIT-III

5. Physico-chemical characteristics of marine environment.

6. *Zonation in marine habitat :* Pelagic and benthic zone and their flora and fauna.

UNIT-IV

7. *Productivity :* Concepts of productivity, primary, secondary and tertiary productivity, yield and carrying capacity.

8. *Ecological succession :* Succession, kinds of succession, changes in animal life during hydrarch, concept of climax.

9. Adaptations of aquatic organisms.

Suggested Books :

1. Ruttner, F., *Fundamentals of Limnology*, University of Tronto Press, Tronto, 1975.

2. Reid, G.K., and Wood, R.D., *Ecology of Inland Waters and Estuaries*, Van Nostrand Reinhold Publishing Co., New York, 1975.

3. Welch, P.S., *Limnology*, McGraw Hill Book Co., New York,1952.

4. Odum, E.P., *Fundamentals of Ecology*, Saunders Co., Philadelphia, 2004.

5. Jhingran, V.G., *Fish & Fisheries of India*, Hindustan Publications Corporation, Delhi, 1991.

Practicals :

Practicals based on both theory papers :

I Estimation of mitotic index/comparison of mitotic indices after a specific treatment by chemical or drug (EDTA).

II Demonstration of male and female reproductive systems and their associated glands in rat/rabbit.

III Histology of testis, epididymis, vas deferens, prostate, seminal vesicle, ovary, uterus and vagina.

IV Study of permanent preparations of the vaginal smear of human menstrual cycle and rat/mice oestrous cycle.

V Determination of the different phases of the oestrous cycle in living rat.

VI Study of mammalian sperm for sperm motility and sperm structure.

VII Determination of sperm count in a mammal with haemocytometer.

VIII Histological examination of placenta.

XI Study of components of lake ecosystem and grazing type of food chain.

X Physico-chemical characteristics of freshwater environment. Temperature, Light, turbidity, current of water, density, pH, dissolved O_2, CO_2 and nutrients.

XI Freshwater zonation in lake/stream and study of different communities (Plankton, nekton, neuston, benthos and lotic communities).

XII Ecological succession in an aquatic habitat in the laboratory.

Scheme for Conduct of Practical Examination :

Max. Marks	:	50
Practical	:	45 marks
Internal Assessment	:	5

Note : 1. Practical will be of 4 hrs. duration.

2. The students will be required to submit in writing the principles, methods employed and precautions to be observed wherever necessary within 1 hr. of the start of examination.

I Dissect and demonstrate the male/female reproductive system of rabbit/rat. Draw a neat and labelled diagram and show it to the examiner. (7)

II	Determine the phase of oestrous cycle in the animal provided. Draw a neat and labelled diagram and show it to the examiner.	(3)
III	Determine the sperm count with haemocytometer in the animal provided. Write the procedure followed.	(4)
IV	Determine following physico-chemical factors in a given sample of fresh water. Write the procedure followed.	(6)
V	Record biota in a given sample from a lake. Make labelled, sketches and demonstrate them to the examiner.	(4)
VI	Examine the water sample for various communities from different zones of a lake and draw labelled sketches.	(4)
VII	Identify the specimens/slides A, B, C & D. Give reasons of the identification.	(1.5 × 4 = 6)
VIII	Note book and charts.	(3+3 = 6)
IX	Viva-Voce.	(5)

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